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Sentinel Flux Core: Modular SAR Response Toolkit



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Declaration

We have read the project guidelines and we understand the meaning of academic dishonesty, in particular plagiarism and collusion. We hereby declare that the work submitted for the final year project, entitled Sentinel Flux Core: Modular SAR Response Toolkit, is original work and has not been printed, published, or submitted before as a final year project, research work, publication, or any other documentation.

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Statement of Submission

This is to certify that the students listed below have successfully submitted the final project named Sentinel Flux Core: Modular SAR Response Toolkit, at the Computer Science & IT Department, The University of Lahore, Lahore, Pakistan, to fulfill the partial requirement of the degree of BS in Computer Science.

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Dedication

This project is dedicated to communities affected by floods, to emergency responders who work under dangerous conditions, and to researchers and public agencies building more reliable disaster-risk intelligence for Pakistan.

Acknowledgement

We acknowledge the open geospatial and remote-sensing communities whose tools and data make this work possible, including Sentinel-1 SAR data ecosystems, Copernicus Emergency Management Service reference products, ESA SNAP processing workflows, scientific Python libraries, and disaster-risk research. We also acknowledge that real operational use requires hydrology, remote-sensing, and disaster-management experts.

Date:

Abstract

Sentinel Flux Core: Modular SAR Response Toolkit is a modular SAR response toolkit for real-data flood detection, benchmark evaluation, machine-learning refinement, and disaster-impact intelligence. The project addresses the problem that flood mapping during monsoon conditions is slow and uncertain when optical imagery is blocked by clouds, when reference products have different footprints, and when pixel maps are not linked to actionable risk evidence. The system uses Sentinel-1 SAR, event manifests, RTC/SNAP-grade preprocessing assumptions, flood-specific detectors, held-out Pakistan AOI benchmarks, calibration metrics, and impact-reporting workflows. The implementation focuses on Pakistan 2022 flood observation as the flagship benchmark domain. It produces reproducible flood probability and mask products, evaluates them with precision, recall, F1, IoU, AP/PR-AUC, ROC-AUC, Brier score, and calibration evidence, and records development targets such as official-reference transfer, preprocessing provenance, and operational validation. The project is an evidence-backed decision-support and research platform with uncertainty-aware, human-reviewed life-safety use.

Area of the Project

Remote sensing, SAR image processing, geospatial artificial intelligence, flood detection, disaster-risk analytics, benchmark engineering, machine-learning refinement, and decision-support software for Pakistan water-risk intelligence.

Technologies used

Python 3.10+, Typer, Rich, NumPy, SciPy, Rasterio, GDAL-compatible GeoTIFF/COG workflows, scikit-learn, PyTorch, PyYAML, Requests, Matplotlib, Pillow, Sentinel-1 SAR, Copernicus Data Space STAC/OData, ESA SNAP GPT concepts, CEMS/GFM reference products, event YAML manifests, JSON provenance, pytest, Ruff, Mermaid diagram rendering, Word report generation, and PDF layout verification.

Acronyms and Abbreviations

- AOI: Area of Interest
- AP: Average Precision
- ARD: Analysis Ready Data
- BFAST: Breaks For Additive Season and Trend
- *CEMS: Copernicus Emergency Management Service*
- *CDSE: Copernicus Data Space Ecosystem*
- *COG: Cloud Optimized GeoTIFF*
- CRS: Coordinate Reference System
- DEM: Digital Elevation Model
- ECE: Expected Calibration Error
- F1: Harmonic mean of precision and recall
- FFD: Flood Forecasting Division
- GFM: Global Flood Monitoring
- GLRT: Generalized Likelihood Ratio Test
- GLOF: Glacial Lake Outburst Flood
- HAND: Height Above Nearest Drainage
- IoU: Intersection over Union
- ML: Machine Learning
- NDMA: National Disaster Management Authority
- OData: Open Data Protocol
- PMD: Pakistan Meteorological Department
- PR-AUC: Precision-Recall Area Under the Curve
- RTC: Radiometric Terrain Correction
- SAR: Synthetic Aperture Radar
- SNAP: Sentinel Application Platform
- STAC: SpatioTemporal Asset Catalog
- VV/VH: Sentinel-1 co-polarized and cross-polarized channels

Table of Content

Declaration ... 1
Statement of Submission ... 2
Dedication ... 3
Acknowledgement ... 4
Abstract ... 5
Area of the Project ... 6
Technologies used ... 6
Acronyms and Abbreviations ... 7
Table of Content ... 8
List of Figures ... 10
List of Tables ... 12
Chapter 1 Introduction ... 14
1.1. Pakistan Flood-Risk Context ... 14
1.2. Safety-Critical Doctrine ... 15
1.3. Problem Statement ... 15
1.4. Objectives ... 15
1.5. Research Questions ... 15
1.6. Scope and Delimitations ... 15
1.7. Significance ... 16
Chapter 2 Literature Review ... 17
2.1. Literature Review Strategy ... 17
2.1.1. Databases Searched ... 17
2.1.2. Keywords ... 17
2.1.3. Inclusion and Exclusion Criteria ... 17
2.1.4. Summary of Articles and Systems ... 17
2.2. Research Evidence Base ... 17
2.3. SAR Change Detection Foundations ... 18
2.4. Operational Flood Mapping Evidence ... 18
2.5. Deep Learning, Uncertainty, and Calibration Evidence ... 19
2.6. Review of Existing Work ... 19
2.7. Comparative Analysis ... 20
2.8. Identified Gap ... 20
2.9. Proposed Solution ... 20
2.10. Conceptual Framework ... 20
2.11. Critical Design Lessons from the Review ... 21
Chapter 3 Requirements and Analysis ... 23
3.1. System Environment and Interfaces ... 23
3.2. Constraints and Assumptions ... 23
3.3. Functional Requirements ... 23
3.4. Non-Functional Requirements ... 24
3.5. Use Case Analysis ... 24
3.5.1. System Level Use Case ... 25
3.5.2. Functional Use Case ... 26
3.5.3. Detailed Functional Use Cases ... 27
3.6. Research Design ... 32
3.7. Data Collection ... 32
3.8. Evaluation Metrics ... 32
Chapter 4 System Design ... 33
4.1. System Architecture Diagram ... 33
4.2. Entity-Relationship Diagram ... 35
4.3. Class Diagram ... 36
4.4. Sequence Diagrams ... 36
4.5. Activity Diagram ... 38
4.6. Data Flow Diagrams ... 48
4.7. Flood Detector Semantics ... 56
4.8. Advanced Flood Probability Model ... 56
4.9. Feature-Based ML Classifier ... 56

4.10. Deep Learning Refinement 57
4.11. Mathematical Formulation 57
Chapter 5 Implementation 58
5.1. Development Environment 58
5.2. Technology Stack 58
5.3. Module Description 58
5.4. Implementation Coverage Audit 58
5.5. Code Evidence Screenshots 59
5.6. User Interface and Implementation Evidence 63
5.7. Command-Line Interface 63
5.8. Output Standardization 63
Chapter 6 Testing and Validation 65
6.1. Unit Testing 65
6.2. Integration Testing 65
6.3. Quality Testing 65
6.4. Local Verification Evidence 65
6.5. Real-Data Benchmark Validation 67
Chapter 7 User Manual 68
7.1. Initial Setup 68
7.2. Validate an Event 68
7.3. Run Flood Detection 68
7.4. Run a Benchmark Suite 68
7.5. Train and Apply the Feature Model 68
7.6. Generate an Impact Report 68
7.7. Dashboard Review 68
Chapter 8 Results and Discussion 70
8.1. Descriptive Statistics 70
8.2. Hypothesis Testing Results 70
8.3. Model Performance Results 70
8.4. Interpretation of Results 70
8.5. Comparison with Literature 70
8.6. Theoretical Implications 71
8.7. Practical Implications 71
Chapter 9 Conclusion and Development Roadmap 72
9.1. Development Roadmap Priorities 72
9.2. Expert-Grade Acceptance Criteria 72
Chapter 10 Advanced Pakistan Operational Design 73
10.1. National Water-Risk Intelligence Scope 73
10.2. Data Reliability Tiers 73
10.3. Operational Product Layers 74
10.4. Forecasting Integration Strategy 74
10.5. ML Governance and Model Cards 74
10.6. Security and Misuse Controls 75
10.7. Deployment Roadmap 75
Chapter 11 References 76
Appendix A Artifact and Verification Map 78
Appendix B Deep Technical Specification 78
B.1. Event Manifest Contract 78
B.2. Raster Semantics Contract 79
B.3. Flood Probability Algorithm Contract 79
B.4. ML Training and Promotion Protocol 79
B.5. Failure Mode Register 80

List of Figures

Figure 2.1. Conceptual model for evidence-backed SAR flood intelligence....21
Figure 3.1. System-level use case diagram....26
Figure 3.2. Benchmark-suite functional use case diagram....27
Figure 3.3. Environment and event validation use case diagram....28
Figure 3.4. Flood detection use case diagram....29
Figure 3.5. ML refinement and model promotion use case diagram....30
Figure 3.6. Impact reporting and dashboard review use case diagram....31
Figure 4.1. System architecture diagram....34
Figure 4.2. Entity-relationship diagram....35
Figure 4.3. Class diagram....36
Figure 4.4.1. Sequence diagram for flood detection....37
Figure 4.4.2. Sequence diagram for feature-model promotion....37
Figure 4.4.3. Sequence diagram for benchmark-suite execution....38
Figure 4.4.4. Sequence diagram for impact and dashboard review....38
Figure 4.5. Activity diagram for event validation and detection....39
Figure 4.5.1. Activity diagram for flood detection workflow....41
Figure 4.5.2. Activity diagram for ML refinement and model promotion....43
Figure 4.5.3. Activity diagram for benchmark-suite execution....45
Figure 4.5.4. Activity diagram for impact report and dashboard review....47
Figure 4.6. DFD Level 0....48
Figure 4.7. DFD Level 1....49
Figure 4.8.1. DFD Level 2 for flood detection....51
Figure 4.8.2. DFD Level 2 for benchmark and ML promotion....53
Figure 4.8.3. DFD Level 2 for impact and dashboard review....55
Figure 5.1. CLI runtime checks and data-validation commands....59
Figure 5.2. Detector semantics and validation domains....60
Figure 5.3. Advanced flood probability from SAR evidence....60
Figure 5.4. Feature-based flood classifier input schema....61
Figure 5.5. Benchmark gates and reference-quality policy....61
Figure 5.6. Event manifest dataclasses for reproducible workflows....62
Figure 5.7. SNAP GPT preprocessing graph construction....62
Figure 5.8. Dashboard raster preview downsampling....63
Figure 7.1. Operational user workflow....69

List of Tables

Table 1.1. Pakistan Hazard Regimes and System Implications....14
Table 2.1. Verified Research Evidence Base....18
Table 2.2. External Evidence and How It Shapes the Project....19
Table 1. Comparative Analysis of Existing Work....20
Table 2. System Environment and Interfaces....23
Table 3.1. System-Level Use Case Details....26
Table 3.2. Benchmark-Suite Use Case Details....27
Table 3.3. Environment and Event Validation Use Case Details....28
Table 3.4. Flood Detection Use Case Details....29
Table 3.5. ML Refinement and Model Promotion Use Case Details....30
Table 3.6. Impact Reporting and Dashboard Review Use Case Details....31
Table 5.4. Research Controls Against Overstated Accuracy....32
Table 6. Evaluation Metrics....32
Table 6.1. Detector Semantics and Scientific Interpretation....56
Table 6.2. Flood Feature Families....56
Table 7. Technology Stack....58
Table 8. Implementation Modules....58
Table 8.1. Implementation Coverage Audit....58
Table 9. Important sarcd Commands....63
Table 10. Unit Test Coverage Summary....65
Table 11. Integration Test Cases....65
Table 11.1. Local Verification Status....66
Table 12. Pakistan Real-Data Benchmark Summary....67
Table 13. Research Question Outcomes....70
Table 14. Key Model Performance Results....70
Table 14.1. Expert-Grade Acceptance Gates....72
Table 15. HydroTwin Pakistan Module Architecture....73
Table 16. Data Reliability Tiers....73
Table 17. Required Product Layers....74
Table 18. Forecasting Maturity Levels....74
Table 19. Required Model Card Fields....74
Table 20. Deployment Roadmap....75
Table 21. Documentation Artifact Map....78
Table 22. Event Manifest Technical Contract....78
Table 23. Raster Metadata Required for Production Products....79
Table 24. Flood Probability Processing Steps....79
Table 25. ML Promotion Checklist....79
Table 26. Failure Mode Register....80

Chapter 1: Introduction

Flood disasters require fast and trustworthy geospatial information. Pakistan is especially exposed to large riverine floods, flash floods, urban drainage failures, and post-flood standing-water impacts. During major monsoon events, cloud cover makes optical satellite mapping unreliable exactly when emergency teams need the clearest picture. Sentinel-1 Synthetic Aperture Radar addresses this limitation because radar can observe the surface through clouds and at night.

Sentinel Flux Core: Modular SAR Response Toolkit was built to turn real Sentinel-1-derived data into reproducible flood products. It includes ingestion helpers, production-oriented preprocessing interfaces, event manifests, flood detectors, machine-learning refinement, benchmark suites, probability calibration, impact reports, and a local dashboard. The repository now focuses on Pakistan 2022 flood mapping as the first serious benchmark before broader forecasting or national-scale operational deployment.

The project operates as decision-support software rather than an automated warning authority. A flood probability layer supports analysts, while official public warnings require agency review, local context, and field validation.

Pakistan Flood-Risk Context

Pakistan includes multiple flood regimes. The lower Indus floodplain, Punjab river corridors, Karachi and Lahore urban drainage systems, Potohar and Balochistan flash-flood catchments, and northern GLOF valleys each fail in different ways. A single satellite segmentation model cannot safely cover all of these regimes without explicit metadata, local references, and hazard-specific assumptions. The 2022 floods provide the first national-scale benchmark because they produced extensive SAR-observable inundation, independent rapid mapping references, severe cropland exposure, and strong relevance to real disaster response.

External evidence confirms the scale of the problem. The World Bank reported more than USD 14.9 billion in damages and about USD 15.2 billion in economic losses from the 2022 Pakistan floods, with reconstruction needs above USD 16 billion. Sentinel-1 research on the same event showed that automated SAR flood mapping can detect tens of thousands of square kilometres of inundation, but also that timing differences and reference-product definitions can change measured accuracy. This is why the project treats benchmark design and reference-quality audit as core scientific work.

Table 1.1. Pakistan Hazard Regimes and System Implications

Hazard Regime	Representative Areas	Detection Need	Operational Need
Riverine Indus flooding	Sindh, south Punjab, lower Indus corridor	Multi-date SAR extent, flood duration, cropland exposure	Evacuation priority, relief routing, crop-loss evidence
Punjab river corridors	Chenab, Ravi, Sutlej, Jhelum, Indus reaches	Reach-specific floodplain mapping and upstream timing context	Barrage stress, embankment inspection, village ranking
Urban pluvial flooding	Karachi, Lahore, Rawalpindi, Hyderabad, Peshawar	High-resolution drainage and road waterlogging evidence	Underpass danger, pump deployment, road closures
Flash floods and hill torrents	KP, Balochistan, DG Khan, Potohar, GB	Rapid rainfall thresholds and terrain susceptibility	Short-lead warning, route isolation, local verification
GLOF risk	Gilgit-Baltistan and KP valleys	Lake inventory, seasonal lake growth, downstream exposure	Preparedness zones, sensor placement, evacuation-time analysis
Post-flood water and health	Sindh, Balochistan, low-drainage floodplains	Flood duration and stagnant-water persistence	WASH, disease-risk, shelter and clinic access planning

Safety-Critical Doctrine

The system is designed around a safety principle suited to probabilistic disaster intelligence: no silent failure, no unsupported confidence, no hidden reference mismatch, no single point of evidence, and no automated life-critical order. The dashboard and reports explain source freshness, model agreement, uncertainty, valid footprint, and known limitations beside every high-impact product.

- Never issue public evacuation orders automatically; the product supports authorized reviewers.
- Never present a probability without calibration evidence or a warning that calibration is unavailable.
- Never evaluate outside a declared valid reference footprint.
- Never hide disagreement between official observations, SAR products, forecast models, and field reports.
- Never promote a model using random tiles alone; promotion requires event/geography holdout.
- Never treat missing data as normal data; missing sources lower confidence and appear in provenance.

Problem Statement

The problem addressed by this project is the lack of a reproducible, real-data, uncertainty-aware SAR flood detection and benchmark workflow that can produce validated Pakistan flood products and connect them to impact intelligence.

Objectives

- Create a reproducible event-manifest workflow for Sentinel-1 flood benchmarks.
- Use RTC/SNAP-grade preprocessing assumptions and mark non-production shortcuts clearly.
- Implement flood-specific detection using dual polarization, temporal anomaly features, terrain masks, and calibrated probabilities.
- Train ML refinement only under strict event/geography holdout validation.
- Evaluate with F1, IoU, precision, recall, AP, ROC-AUC, Brier score, calibration error, and false-positive/false-negative areas.
- Generate benchmark leaderboards for Pakistan AOIs and expose weak events clearly.
- Convert flood pixels into impact-oriented outputs for roads, built-up areas, facilities, and other exposed assets.
- Provide CLI and dashboard interfaces for reproducible local operation.
- Maintain a safety boundary: the system recommends and documents evidence, while official agencies issue warnings.

Research Questions

- RQ1: Can Sentinel-1 RTC/ARD data and reference products produce a Pakistan flood detector that generalizes across multiple AOIs?
- RQ2: Does a feature-based flood classifier improve held-out AOI performance compared with a single threshold detector?
- RQ3: Which benchmark split exposes real generalization failures more reliably: random tiles or event/geography holdout?
- RQ4: Can probability calibration and uncertainty reporting make flood maps more useful for operational review?
- RQ5: What additional impact layers are required to convert a flood mask into actionable decision-support evidence?

Scope and Delimitations

The project scope includes local real-data flood detection, Pakistan benchmark suites, reference mask staging, feature-based ML refinement, probability calibration, impact reporting, and dashboard manifest

generation. The strongest implemented workflow is observational flood mapping, not autonomous prediction.

- Included: Sentinel-1 data discovery/staging, event validation, flood detection, ML refinement, evaluation, calibration, impact reports, and dashboard manifests.
- Included: Pakistan 2022 AOIs using GFM, CEMS, and UNOSAT/VIIRS-style reference comparisons where staged locally.
- Excluded from operational readiness: automated evacuation orders, official public warnings, and life-critical decisions without human authority.
- Outside the present acceptance gate: full hydrodynamic forecasting, city-scale SWMM deployment, and GLOF exact-timing prediction.
- Delimitation: final accuracy depends on reference quality, SAR preprocessing quality, valid-mask footprints, and AOI representativeness.

Significance

The significance of the project is practical and scientific. Practically, it can reduce the time needed to produce flood extent and exposure evidence from SAR data. Scientifically, it establishes a strict benchmark workflow that makes it harder to overstate model quality. The project reports both strong clean-AOI results and weaker audit-AOI results, which is essential for trustworthy disaster-risk systems.

Chapter 2: Literature Review and Existing Work

Flood mapping systems generally combine satellite observations, hydrological context, reference water masks, terrain constraints, and validation data. The key lesson from existing operational systems is that no single model is sufficient for high-stakes use. SAR flood products are strongest when they are backed by quality preprocessing, multiple algorithms, exclusion masks, uncertainty, and independent validation.

Literature Review Strategy

The review emphasized official data systems, remote-sensing flood mapping literature, and operational disaster-management workflows. The repository plan was grounded in Sentinel-1 SAR, Copernicus data access, ESA SNAP processing, CEMS Global Flood Monitoring design patterns, and real Pakistan 2022 flood references.

Databases Searched

- Copernicus Data Space Ecosystem documentation for STAC and OData access.
- ESA SNAP Microwave Toolbox and Sentinel-1 processing references.
- Copernicus Emergency Management Service and openEO GFM workflow descriptions.
- NASA and Earthdata SAR processing recipes for radiometric terrain correction.
- Project repository documents under docs/ and benchmark artifacts under data/benchmarks/.
- Remote-sensing research on SAR flood detection, temporal change detection, and uncertainty-aware evaluation.

Keywords

Sentinel-1, SAR flood mapping, RTC, SNAP GPT, Global Flood Monitoring, Pakistan 2022 floods, change detection, log ratio, GLRT, dual polarization, HAND, calibration, event holdout, benchmark atlas, impact analysis, COG, disaster decision support.

Inclusion and Exclusion Criteria

- Included sources that describe operational SAR preprocessing, benchmark reference products, or validated flood-mapping workflows.
- Included implementation evidence from repository source files, test files, configs, and generated benchmark JSON.
- Excluded unsupported claims of perfect flood prediction or zero failure.
- Excluded random-tile-only validation as a final expert metric because it can leak geography and event characteristics.
- Excluded raw Sentinel-1 measurement shortcuts as production science outputs unless explicitly marked as development quicklooks.

Summary of Articles and Systems

The reviewed operational pattern is consistent: ingest trustworthy data, perform physics-aware preprocessing, apply multiple flood indicators, use exclusion masks, produce likelihood or probability outputs, evaluate against independent references, and show uncertainty. The toolkit mirrors this pattern in a local Python workflow and uses Pakistan AOIs as the first rigorous real-data proving ground.

Research Evidence Base

The research foundation combines peer-reviewed SAR change-detection methods, operational flood-mapping systems, probability-calibration research, and official Sentinel-1 data-access documentation. The

selected sources support the project decisions that matter most: scale-aware SAR preprocessing, explicit detector semantics, spatially adaptive flood evidence, reference-footprint validation, probability calibration, and held-out Pakistan benchmark evaluation.

Table 2.1. Verified Research Evidence Base

Evidence Source	Relevant Contribution	Design Implication
Bovolo and Bruzzone, IEEE TGRS 2005	Multitemporal SAR change detection using scale-aware, detail-preserving methods	Background for log-ratio and multiscale SAR change detection
Conradsen et al., IEEE TGRS 2003	Wishart-distribution test statistic for polarimetric SAR change detection	Statistical basis for GLRT-style change evidence
Lee, IEEE PAMI 1980	Adaptive local-statistics speckle filtering	Establishes speckle control as a preprocessing requirement
Schumann and Moller, Physics and Chemistry of the Earth 2015	Microwave remote sensing review for flood inundation mapping	Supports SAR use for cloudy flood conditions
Martinis, Twele, and Voigt, NHESS 2009	Split-based near-real-time flood thresholding on high-resolution SAR	Operational thresholding precedent
Twele, Cao, Plank, and Martinis, IJRS 2016	Fully automated Sentinel-1 flood mapping chain	Sentinel-1 flood product design precedent
Chini et al., IEEE TGRS 2017	Hierarchical split-based parametric SAR thresholding	Shows need for spatially adaptive thresholds
Mondini et al., Earth-Science Reviews 2021	Review of SAR landslide mapping limits and methods	Landslide roadmap context
Aimaiti et al., Remote Sensing 2019	PALSAR-2 amplitude-change landslide mapping after Hokkaido earthquake	Landslide detector roadmap context
Gao et al., IEEE GRSL 2016	PCANet-based SAR change detection	ML change-detection background
Daudt, Le Saux, and Boulch, ICIP 2018	Fully convolutional Siamese networks for change detection	Paired-image neural change detection background
Bonafilia et al., CVPRW 2020	Sen1Floods11 Sentinel-1 flood dataset	External ML training benchmark paired with Pakistan holdout validation
Gal and Ghahramani, ICML 2016	Dropout as approximate Bayesian inference	Basis for MC-dropout uncertainty when neural refinement is used
Platt 1999; Zadrozny and Elkan 2002; Guo et al. 2017	Post-hoc probability calibration methods and neural calibration risks	Supports Brier score, ECE, reliability bins, and calibration policy
Roth et al., NHESS 2023 Pakistan flood analysis	Sentinel-1 analysis of Pakistan 2022 floods and large-scale automated mapping feasibility	Pakistan 2022 flood benchmark grounding

SAR Change Detection Foundations

SAR backscatter is affected by surface roughness, moisture, viewing geometry, polarization, terrain, and speckle. Classical change detection therefore compares pre-event and post-event observations through ratios, differences, likelihood-ratio statistics, or temporal anomalies rather than interpreting a single image in isolation. The retained Bovolo and Bruzzone work supports the basic idea that multitemporal SAR change requires transformations that preserve real spatial detail while suppressing noise. In the toolkit, that lesson appears as detector semantics: data_scale is explicit, change_direction is explicit, and probability outputs are separated from binary decisions.

The GLRT literature is useful because it formalizes change detection as a hypothesis test, while the full polarimetric Wishart formulation requires product types beyond every Sentinel-1 GRD/RTC product used in this project. GLRT-style evidence is therefore used as a statistical design influence while the Pakistan benchmark remains based on the available Sentinel-1 product semantics. That distinction is important for scientific validity.

Speckle filtering remains relevant, and production preprocessing uses SNAP GPT, ASF HyP3 RTC, OPERA RTC-S1, Planetary Computer RTC, or an equivalent validated RTC/ARD source. Python-only preprocessing is retained as a development path separate from the final scientific default.

Operational Flood Mapping Evidence

The verified flood-mapping literature supports a conservative architecture: preprocessing, spatially adaptive flood evidence, water/reference masks, terrain exclusions, and independent validation. Martinis et al. and Twele et al. show why fully automated SAR flood mapping requires more than one global threshold. Chini et al. further supports spatially adaptive thresholding in heterogeneous scenes. The project implements the same principle locally through dry-baseline features, dual-polarization evidence, context features, valid footprints, and benchmarked threshold policies.

The Pakistan 2022 literature and CEMS/GFM product architecture are especially important for this project. They show that automated Sentinel-1 flood mapping is feasible over Pakistan but that accuracy depends on timing, reference definition, preprocessing, and valid footprints. The validation design therefore centers held-out AOI evaluation, worst-event performance, and reference-quality audit instead of treating one clean aggregate score as proof of operational readiness.

Deep Learning, Uncertainty, and Calibration Evidence

PCANet, Siamese fully convolutional networks, Sen1Floods11, MC dropout, and probability calibration are relevant to the ML layer, but ML remains a refinement stage rather than the first dependency. A model is useful only if it improves held-out AOI performance, reports calibration quality, and exposes failure cases. Sen1Floods11 is valuable as an external flood dataset, but Pakistan operational claims still require Pakistan AOI holdouts and local reference audit.

Calibration is a core safety property. In a disaster-risk product, probability represents observed frequency. Platt scaling, isotonic regression, and temperature scaling provide different ways to map model scores into better calibrated probabilities; the implemented project emphasizes monotonic calibration and reports Brier score and ECE where references exist.

Table 2.2. External Evidence and How It Shapes the Project

Evidence Source	Relevant Finding	Design Consequence
CDSE STAC/OData documentation	Sentinel products can be searched through catalog APIs and downloaded through authenticated product APIs	Separate search planning from download/staging and preserve product identifiers
ESA SNAP Microwave Toolbox	Supports calibration, speckle filtering, coregistration, orthorectification, polarimetry, interferometry, and terrain correction	Treat SNAP/RTC as production preprocessing rather than ad hoc raw-raster math
openEO/CEMS GFM	GFM combines Sentinel-1 preprocessing, multiple flood algorithms, ensemble output layers, reference water, exclusions, and likelihood values	Model local product architecture after ensemble, masks, and likelihood/probability products
NHESS Pakistan 2022 Sentinel-1 study	Automated SAR mapping of Pakistan 2022 floods is feasible but accuracy depends on timing and reference comparison	Use multi-AOI benchmarks and reference-quality audit rather than one scene score
World Bank Pakistan 2022 assessment	Flood damages exceeded USD 14.9 billion with about USD 15.2 billion economic losses	Prioritize impact and exposure layers alongside flood pixels
WMO early-warning evidence	Countries with weaker MHEWS have much higher disaster mortality ratios	Design outputs for warning ecosystems while keeping official authority boundaries
UNDP GLOF-II Pakistan	Northern Pakistan has thousands of glacial lakes and dozens of hazardous lakes	Keep GLOFWatch as a later module with risk-based, not exact-timing, claims
GloFAS and Google Flood Forecasting	Riverine forecasts can provide multi-day probabilistic context	Add forecast adapters only after observational flood detection is validated

Review of Existing Work

Copernicus Data Space provides modern access to Sentinel data through STAC search and OData download flows. ESA SNAP supports orbit correction, calibration, speckle filtering, terrain correction, and terrain flattening, which are critical for reliable Sentinel-1 backscatter products. CEMS GFM provides a strong product pattern: Sentinel-1 IW GRDH, backscatter preprocessing, multiple algorithms, reference water, exclusion masks, and likelihood outputs. The project adopts these ideas while keeping local reproducibility and benchmark reporting as first-class engineering outputs.

Classical SAR flood detection often uses low backscatter after flooding, log-ratio change, normalized difference, or likelihood-ratio statistics. These are transparent but sensitive to land cover, permanent water, terrain, wind roughening, building layover, agricultural seasonality, and preprocessing differences. The implemented flood-feature classifier therefore combines engineered SAR features with context and masks rather than relying on one global threshold.

Comparative Analysis

Table 1. Comparative Analysis of Existing Work

System or Method	Strengths	Limitations	Project Response
Manual flood mapping	Expert interpretation and local judgment	Slow, inconsistent, difficult to reproduce at scale	Automated event manifests and benchmark reports
Simple SAR thresholding	Transparent and fast	Weak transfer across land cover, orbit, and reference footprints	Dual-pol fusion, temporal features, terrain masks, and calibrated probability
CEMS GFM pattern	Operational flood-monitoring design with masks and likelihood outputs	External product access and reference scope may not match every local AOI	Use as reference/comparison and reproduce similar architecture locally
Deep learning flood segmentation	Can learn complex spatial patterns	Requires large validated labels and can overfit geography	Use ML as held-out refinement after the transparent flood baseline
Hydrological forecasting	Adds lead time before water arrives	Needs gauge/rainfall reliability, calibration, and official integration	Kept as later module after observational detection is benchmarked

Identified Gap

- Many prototypes report high accuracy on small or random validation splits but do not prove AOI transfer.
- Many demos produce masks without probability, calibration, valid footprints, or reference-quality audits.
- Few local workflows connect flood extent to impact layers such as roads, built-up areas, and facilities.
- Many systems blur advisory outputs with operational warning authority, creating safety risk.
- Reproducible local event manifests and benchmark leaderboards are often missing.

Proposed Solution

The proposed solution is a reproducible SAR flood detection and benchmarking toolkit. It uses event YAML manifests, validated raster alignment, production-oriented RTC/SNAP preprocessing, flood-specific detection, feature-based ML, strict held-out AOI evaluation, calibrated probability outputs, impact reporting, and dashboard-ready manifests. The first expert milestone is Pakistan flood detection on multiple independent AOIs, followed by impact intelligence and later forecasting integration.

Conceptual Framework

The conceptual model has five layers: evidence ingestion, preprocessing and validation, detection and ML refinement, benchmark and calibration, and decision-support outputs. Figure 2.1 shows this evidence chain and the safety boundary between analytical decision support and official warning authority.

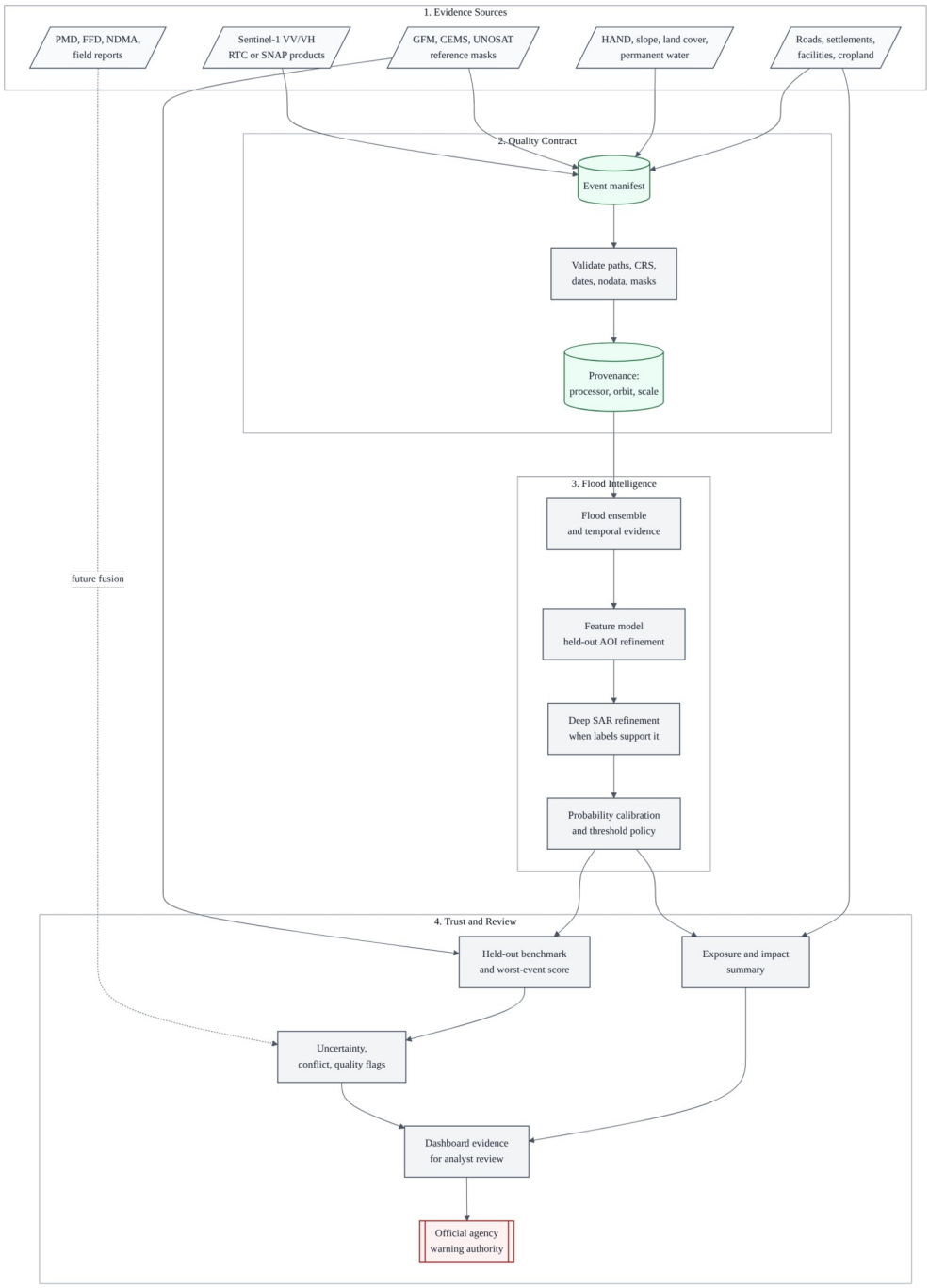


Figure 2.1. Conceptual model for evidence-backed SAR flood intelligence

Critical Design Lessons from the Review

The most important design lesson is that flood intelligence goes beyond image segmentation. A flood pixel is interpreted through observation time, river state, permanent-water history, terrain, land cover, reference

validity, and exposure. A high-scoring segmentation model can still be operationally dangerous if it fails silently in urban layover, vegetated floodplains, or agricultural fields whose backscatter changes for reasons unrelated to floodwater.

The second lesson is that data engineering is part of the science. Sentinel-1 GRD measurement rasters, RTC products, dB-scaled products, sigma0 products, and gamma0 products cannot be mixed without metadata. Orbit direction, relative orbit, acquisition date, polarization, DEM source, terrain correction, speckle filtering, and valid mask all affect downstream thresholds. The repository therefore encodes detector semantics and event manifests instead of relying on free-form path conventions.

The third lesson is that official products and global products function as evidence sources, not blind substitutes for local validation. CEMS GFM is a strong architectural reference because it uses multiple SAR algorithms and supporting layers, but local benchmark suites still need to audit spatial footprints, dates, class definitions, and reference uncertainty. The weaker CEMS-only benchmark in this project is therefore useful: it reveals where model and reference alignment still require expert-grade audit.

Chapter 3: Requirements, Analysis, and Research Methodology

System Environment and Interfaces

Table 2. System Environment and Interfaces

Interface	Purpose	Implementation Evidence
Command line	Run ingestion, validation, detection, benchmark, calibration, impact, and dashboard manifest workflows	Typer app in src/sar_cd_toolkit/cli.py
Event manifests	Define AOI, scenes, products, references, ancillaries, outputs, and provenance	configs/events/*.yaml and src/sar_cd_toolkit/events/config.py
Benchmark suites	Group events and enforce held-out evaluation gates	configs/benchmarks/*.yaml and src/sar_cd_toolkit/benchmark.py
CDSE STAC/OData	Search and download Sentinel products	src/sar_cd_toolkit/ingest/stac.py and odata.py
SNAP GPT	Production-grade Sentinel-1 preprocessing path	src/sar_cd_toolkit/preprocess/ snap_processor.py
Raster IO	Read/write COG/GeoTIFF rasters with metadata	Rasterio utility code and detector pipelines
Dashboard	Review layers and manifests locally	src/sar_cd_toolkit/dashboard/app.py

Constraints and Assumptions

- Production science products use RTC/ARD or SNAP GPT preprocessing; Python-only preprocessing remains development/quicklook unless explicitly validated.
- Reference labels can be imperfect, and poor AOI reference quality is reported separately from aggregate metrics.
- A probability raster is analytical evidence; official warnings require human review and agency authority.
- Large rasters are processed window-by-window or through overviews/tiles to avoid unsafe memory use.
- Pakistan 2022 is the first benchmark domain; national transfer requires additional events and seasons.
- Network credentials, external data entitlements, mounted disks, and SNAP installation are environmental dependencies checked by doctor and validate-data workflows.

Functional Requirements

The product will provide the following key features:

- FR-01: The system shall validate event manifests before producing outputs. Implementation evidence: sarcd validate-data.
- FR-02: The system shall search Sentinel-1 products through STAC. Implementation evidence: sarcd search and sarcd search-gfm.
- FR-03: The system shall stage or reference RTC/SAFE products with provenance. Implementation evidence: ingest and preprocess modules.
- FR-04: The system shall run flood-specific detection with dual polarization where available. Implementation evidence: detect-flood and flood_advanced.py.
- FR-05: The system shall create calibrated probability rasters and binary masks. Implementation evidence: probability_calibration.py and evaluation tools.
- FR-06: The system shall support normalized difference, log ratio, GLRT, and flood ensemble detector semantics. Implementation evidence: detect/config.py and detect/pipeline.py.
- FR-07: The system shall train a feature-based flood model under held-out AOI validation. Implementation evidence: train-flood-feature-model and benchmark suites.
- FR-08: The system shall train and apply neural SAR flood refinement where labeled tiles exist. Implementation evidence: ml/deep_learning.py and ml/pipeline.py.

- FR-09: The system shall compute precision, recall, F1, IoU, AP, ROC-AUC, Brier score, calibration error, and area metrics. Implementation evidence: utils/evaluate.py and benchmark.py.
- FR-10: The system shall generate benchmark leaderboards for event suites. Implementation evidence: benchmark-suite and benchmark-report.
- FR-11: The system shall report reference-quality warnings and valid-footprint constraints. Implementation evidence: events/validation.py and benchmark JSON.
- FR-12: The system shall estimate exposure impacts for supported vector layers. Implementation evidence: impact.py and impact-suite.
- FR-13: The system shall generate dashboard manifests from event outputs. Implementation evidence: dashboard-manifest.
- FR-14: The system shall support no-absolute-path checks for portable configs. Implementation evidence: tests/test_no_absolute_paths.py.
- FR-15: The system shall preserve provenance metadata for output rasters and reports. Implementation evidence: catalog and manifest utilities.

Non-Functional Requirements

The system must satisfy the following quality requirements:

- NFR-01: Reproducibility. Each benchmark result shall identify event manifests, detector, threshold policy, reference, and valid mask.
- NFR-02: Scientific validity. Evaluation shall be restricted to declared valid reference footprints.
- NFR-03: Safety. The product shall label outputs as advisory and shall not automate public evacuation orders.
- NFR-04: Accuracy reporting. Aggregate metrics shall include worst-event performance alongside macro average.
- NFR-05: Calibration. Probability outputs shall be evaluated with Brier score and expected calibration error where references exist.
- NFR-06: Scalability. Raster processing shall avoid loading full national-scale rasters where tiled or overview processing is possible.
- NFR-07: Portability. Configuration files shall avoid committed machine-specific absolute paths.
- NFR-08: Maintainability. Core workflows shall be covered by automated tests and a scoped Ruff lint gate.
- NFR-09: Traceability. Every report output shall be traceable to input data, configuration, and model version.
- NFR-10: Extensibility. Forecasting, landslide, and urban modules shall be separable from the flood benchmark core.

Use Case Analysis

The use case model separates engineering users, scientific users, disaster analysts, dashboard reviewers, and official agency reviewers. This separation keeps model development, analytical review, and public warning authority as distinct responsibilities.

System Level Use Case



Figure 3.1. System-level use case diagram

Table 3.1. System-Level Use Case Details

Field	Details
Scope	End-to-end SAR flood-response workflow for Sentinel Flux Core, from event setup and data staging through detection, benchmarking, impact reporting, dashboard review, and official approval.
Primary actors	Data Engineer; SAR Analyst; ML Engineer; Disaster Analyst; Dashboard User; Official Agency Reviewer.
External systems	Copernicus Data Space for Sentinel product discovery and staging. Local runtime services provide file-system storage, validation, and dashboard artifacts.
Trigger	A flood event, benchmark suite, or operational review request requires reproducible SAR flood evidence.
Preconditions	Environment variables and SAR_DATA_ROOT are configured; event YAML or suite YAML exists; credentials and reference layers are available where required.
Main success flow	Configure event manifest; search and stage data; validate data; run flood detection; benchmark AOI suite; train or calibrate refinement models; generate impact report; review dashboard; approve an operational message when evidence is sufficient.
Includes and extensions	Flood detection, benchmark, training, impact, and review paths include data validation. Training extends benchmarking when a candidate model is promoted. Dashboard review includes official approval only after the evidence package is sufficient.
Alternative flows	Missing credentials or invalid data return actionable failures. Weak benchmarks trigger audit or retraining. Insufficient review evidence blocks official messaging and returns the product for additional validation.
Outputs	Staged SAR products, event manifests, probability and binary flood rasters, metrics, model cards, impact reports, dashboard manifests, and official-review evidence packages.
Postconditions	Every generated artifact is traceable to its inputs, configuration, metrics, and review status; public action remains under official agency authority.

Functional Use Case

The major functional use cases are Validate Event, Detect Flood, Benchmark Suite, Train Feature Model, Create Impact Report, and Review Dashboard. The benchmark use case is the most important scientific

gate because it prevents model promotion unless the model succeeds on held-out AOIs and records the reason for failure when it does not.

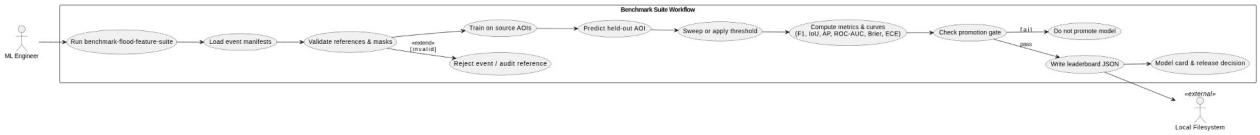


Figure 3.2. Benchmark-suite functional use case diagram

Table 3.2. Benchmark-Suite Use Case Details

Field	Details
Scope	Scientific benchmark and promotion workflow for validating flood detectors and feature models across held-out AOIs.
Primary actor	ML Engineer.
Supporting actors	Evaluation engineer or SAR analyst reviews weak AOIs, reference masks, metrics, and promotion evidence.
External system	Local Filesystem stores event manifests, model artifacts, leaderboard JSON, model cards, and promotion evidence.
Trigger	The engineer runs benchmark-flood-feature-suite or a benchmark-suite command against a declared suite YAML.
Preconditions	Suite YAML identifies source and held-out AOIs; references, masks, and products are staged; validation rules and metric thresholds are declared.
Main success flow	Load event manifests; validate references and masks; train on source AOIs; predict held-out AOI; sweep or apply thresholds; compute F1, IoU, AP, ROC-AUC, Brier, and ECE; check promotion gate; write leaderboard JSON and model-card evidence.
Alternative flows	Invalid references reject the event or require audit. Promotion-gate failure blocks release and records hard negatives for retraining.
Outputs	Per-event metrics, aggregate leaderboard, calibration curves, model card, release decision, and hard-negative evidence.
Acceptance criteria	The candidate model must beat the flood ensemble under the declared holdout policy and satisfy macro and worst-event thresholds.

Detailed Functional Use Cases

The template requires each major functionality to be documented through a use-case description table. The tables below use that structure and bind the user action, system behavior, failure controls, and linked functional requirements to implemented commands.

Figures 3.3 through 3.6 show the workflows where validation, evidence, model promotion, impact reporting, and human review boundaries are enforced.

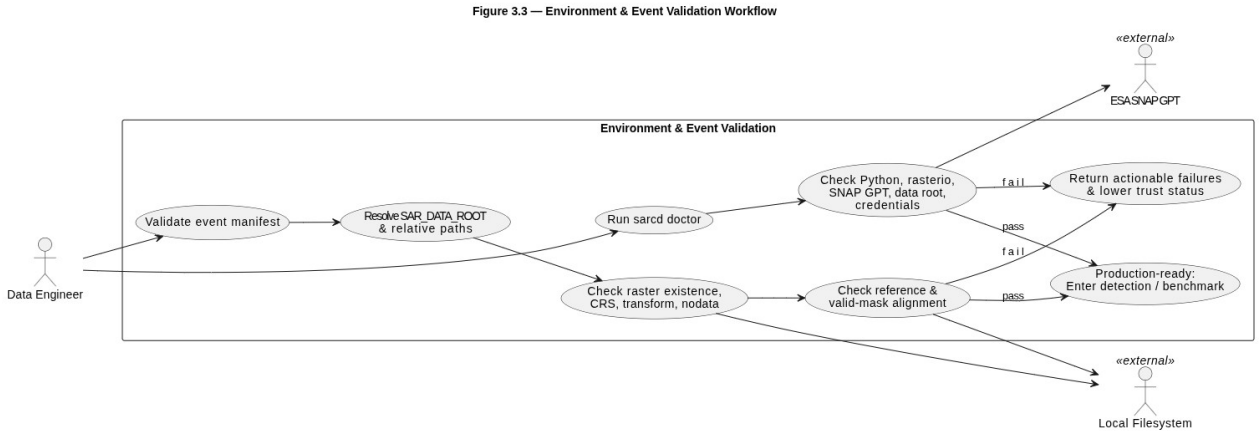


Figure 3.3. Environment and event validation use case diagram

Table 3.3. Environment and Event Validation Use Case Details

Field	Details
Scope	Validation gate that determines whether an event configuration and runtime environment are trustworthy enough for detection or benchmarking.
Primary actor	Data Engineer.
External systems	ESA SNAP GPT, Python geospatial runtime, credential store, event data root, and Local Filesystem.
Trigger	The user validates an event manifest or runs sardc doctor before operational processing.
Preconditions	Event manifest, SAR_DATA_ROOT, expected raster paths, masks, reference products, and optional SNAP configuration are declared.
Main success flow	Validate event manifest; resolve SAR_DATA_ROOT and relative paths; run environment doctor; check Python, rasterio, SNAP GPT, data root, and credentials; verify raster existence, CRS, transforms, nodata, references, and valid-mask alignment; mark the event production-ready.
Alternative flows	Missing dependencies, credentials, rasters, invalid CRS, or mismatched reference masks return actionable failures and lower the trust status.
Outputs	Validation result, actionable failure list, reference-quality status, production-ready flag, and provenance for downstream workflows.
Acceptance criteria	All required paths resolve; rasters align spatially; references and valid masks agree; environment checks pass or have documented non-blocking warnings.

Postconditions	Only validated events enter detection, benchmark, or dashboard workflows; failed events remain blocked until corrected.

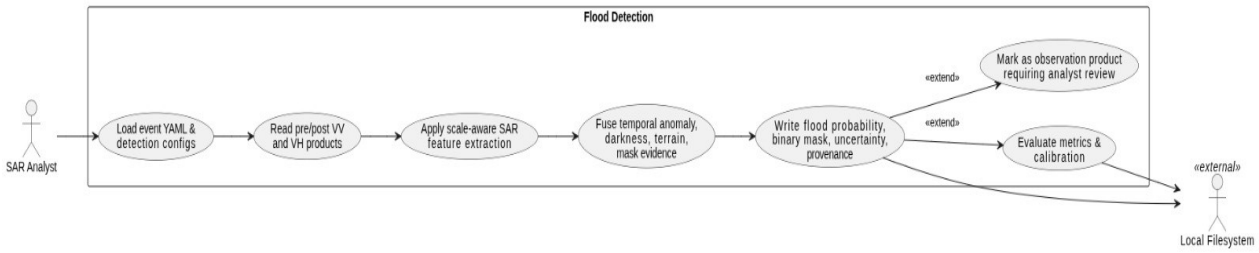


Figure 3.4. Flood detection use case diagram

Table 3.4. Flood Detection Use Case Details

Field	Details
Scope	Operational SAR flood detection workflow that converts validated pre/post SAR products into probability, binary, uncertainty, and provenance outputs.
Primary actor	SAR Analyst.
External system	Local Filesystem stores event YAML, configuration files, source rasters, generated COG/GeoTIFF rasters, JSON provenance, and metrics.
Trigger	The analyst runs sardc detect-flood --event EVENT.yaml or an equivalent detection command.
Preconditions	Event YAML and detection configs are valid; pre/post VV and optionally VH products are available; masks, terrain context, and reference layers are staged where required.
Main success flow	Load event YAML and detection configs; read pre/post VV and VH products; apply scale-aware SAR feature extraction; fuse temporal anomaly, darkness, terrain, and mask evidence; write flood probability, binary mask, uncertainty, and provenance.
Alternative flows	Single-polarisation events use VV-only extraction. Missing reference masks produce an observation product requiring analyst review instead of evaluated-product status.
Outputs	Flood probability raster, binary mask, uncertainty raster, provenance JSON, metrics and calibration summary where references exist.
Acceptance criteria	Outputs are georeferenced, provenance-linked, spatially masked, and either evaluated within a valid reference footprint or clearly marked for analyst review.
Postconditions	Flood products are ready for benchmarking, impact

	reporting, or dashboard review according to their evaluation status.

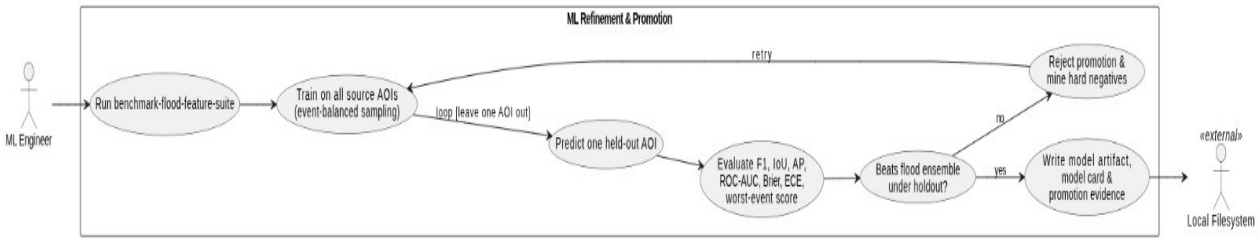


Figure 3.5. ML refinement and model promotion use case diagram

Table 3.5. ML Refinement and Model Promotion Use Case Details

Field	Details
Scope	Model-refinement workflow that trains, validates, rejects, or promotes a feature-based flood model under event-balanced held-out AOI sampling.
Primary actor	ML Engineer.
External system	Local Filesystem stores benchmark data, trained model artifacts, model cards, leaderboard outputs, and promotion evidence.
Trigger	The engineer runs benchmark-flood-feature-suite or a training benchmark over a declared source/holdout AOI suite.
Preconditions	Training AOIs, held-out AOIs, SAR products, HAND, masks, references, feature configuration, and benchmark thresholds are declared and validated.
Main success flow	Train on all source AOIs; leave one AOI out for validation; predict held-out AOI; evaluate F1, IoU, AP, ROC-AUC, Brier, ECE, and worst-event score; compare against flood ensemble; write model artifact, model card, and promotion evidence when the gate passes.
Alternative flows	If the candidate fails the holdout comparison, reject promotion and mine hard negatives such as permanent water edges, wet cropland, or urban shadow before retraining.
Outputs	Trained model artifact, model card, promotion decision, hard-negative set, metrics, calibration evidence, and leaderboard record.
Acceptance criteria	The model improves over the flood ensemble under holdout and meets declared macro and worst-event thresholds with documented calibration.

Postconditions	Promoted models become eligible for operational detection; rejected models remain experimental with recorded failure evidence.

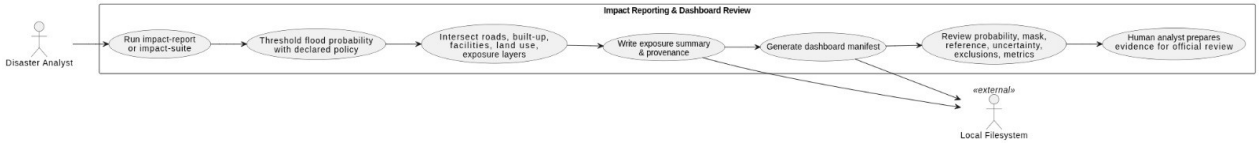


Figure 3.6. Impact reporting and dashboard review use case diagram

Table 3.6. Impact Reporting and Dashboard Review Use Case Details

Field	Details
Scope	Decision-support workflow that turns flood products into exposure summaries, dashboard manifests, and analyst-reviewed evidence packages.
Primary actor	Disaster Analyst.
Supporting actors	Dashboard reviewer and official agency reviewer inspect layers, provenance, uncertainty, exclusions, and metrics before public action.
External system	Local Filesystem stores flood products, exposure layers, summary JSON, dashboard manifests, and review evidence.
Trigger	The analyst runs impact-report, impact-suite, or dashboard-manifest after flood probability or binary products are available.
Preconditions	Flood probability or binary mask exists; threshold policy is declared; roads, built-up areas, facilities, land use, and other exposure vectors are staged and CRS-compatible.
Main success flow	Apply declared threshold policy; harmonise raster/vector CRS; intersect flood mask with exposure layers; summarize affected assets and area; write exposure summary and provenance; generate dashboard manifest; review probability, mask, reference, uncertainty, exclusions, and metrics; prepare official-review evidence.
Alternative flows	Uncertainty, missing exposure layers, invalid CRS, or insufficient evidence trigger additional validation or review flags instead of official approval.
Outputs	Impact JSON and tables, exposure summaries, dashboard manifest, review-ready map layers, provenance, and official-review evidence package.
Acceptance criteria	Dashboard evidence is complete, traceable, uncertainty-aware, and sufficient for a human analyst to recommend official review.

Research Design

The research design is experimental and comparative. The project compares flood detectors and ML refinement under real data constraints. The preferred validation mode is event/geography holdout: train on some AOIs, validate or calibrate on separate AOIs, and test on held-out AOIs. This is stricter than random tile splits because it tests geographic transfer.

Table 5.4. Research Controls Against Overstated Accuracy

Risk	Control in This Project	Evidence Artifact
Random tile leakage	Use event/geography holdout suites	benchmark-suite JSON
Invalid reference footprint	Require valid mask and reference-quality status	validate-data and event report
Ambiguous probability outputs	Require threshold or calibration for evaluation	evaluate and calibration commands
Aggregate hiding failures	Report worst-event F1 and gate results	leaderboard JSON
Machine-specific paths	Resolve through SAR_DATA_ROOT and no-absolute-path checks	manifest tests
Unclear preprocessing	Persist processor, scale, orbit, polarization, DEM, and graph metadata	preprocess and event manifests

Data Collection

- Sentinel-1 scene metadata and product discovery through CDSE STAC/OData and Planetary Computer RTC staging.
- Reference products from GFM and CEMS-derived Pakistan AOIs staged as raster masks and valid footprints.
- UNOSAT/VIIRS-style audit references for coarse independent comparison where staged locally.
- Ancillary layers such as HAND, slope, land cover, permanent-water masks, and vector exposure layers.
- Generated outputs including probability rasters, binary masks, benchmark JSON, calibration reports, and impact reports.

Evaluation Metrics

Table 6. Evaluation Metrics

Metric	Meaning	Why It Matters
Precision	Fraction of predicted flood pixels that are reference flood	Controls false alarms
Recall	Fraction of reference flood pixels detected	Controls missed flooded areas
F1	Harmonic mean of precision and recall	Single balanced detection score
IoU/CSI	Intersection over union between predicted and reference flood	Spatial overlap quality
AP/PR-AUC	Ranking quality under class imbalance	Useful for probability products
ROC-AUC	Separability across thresholds	Shows threshold-independent discrimination
Brier score	Mean squared probability error	Probability calibration quality
ECE	Expected calibration error	Whether stated probabilities match observed frequency
False-positive area	Area of reference dry pixels predicted as flood	Operational over-warning risk
Worst-event F1	Lowest event F1 in suite	Generalization and safety risk indicator

Chapter 4: System Design and Model Development

The toolkit is designed as a modular local pipeline. Data ingestion discovers or stages products; preprocessing converts SAR data into comparable backscatter; event manifests bind inputs and outputs; detection modules create score/probability/binary rasters; benchmark modules evaluate products; impact modules convert flood pixels into exposed assets; dashboard manifests expose results for human review.

System Architecture Diagram

The architecture is a modular pipeline. Each layer writes explicit artifacts so that a benchmark result can be traced back to source data, preprocessing, detector configuration, reference mask, threshold policy, and model version.

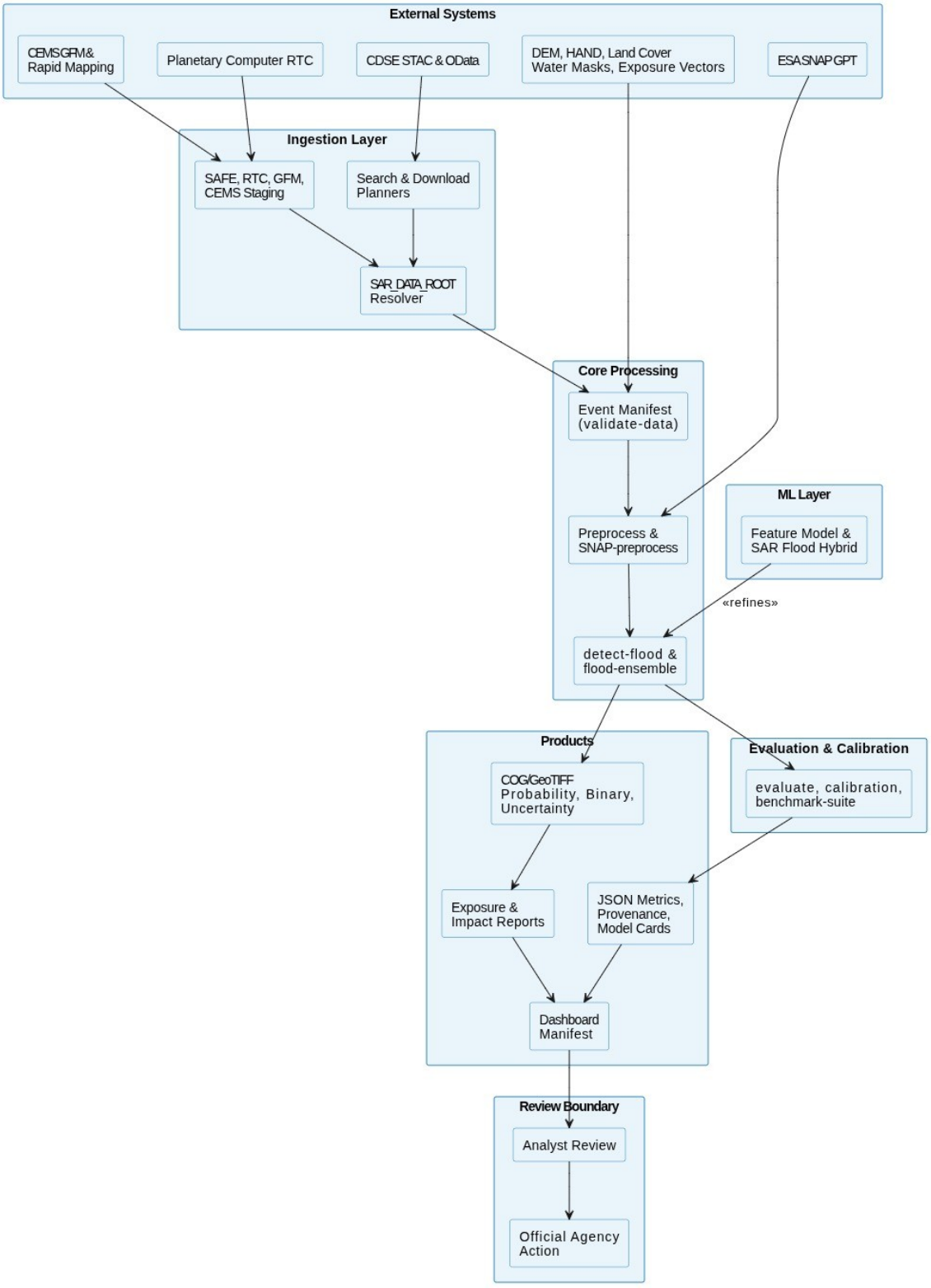


Figure 4.1. System architecture diagram

Entity-Relationship Diagram

The entity model links Event, Scene, PreprocessingProduct, DetectionConfig, ReferenceMask, AncillaryLayer, ModelArtifact, BenchmarkRun, MetricSet, CalibrationFile, ImpactLayer, and DashboardManifest. These entities map directly to YAML/JSON configs and artifact paths.

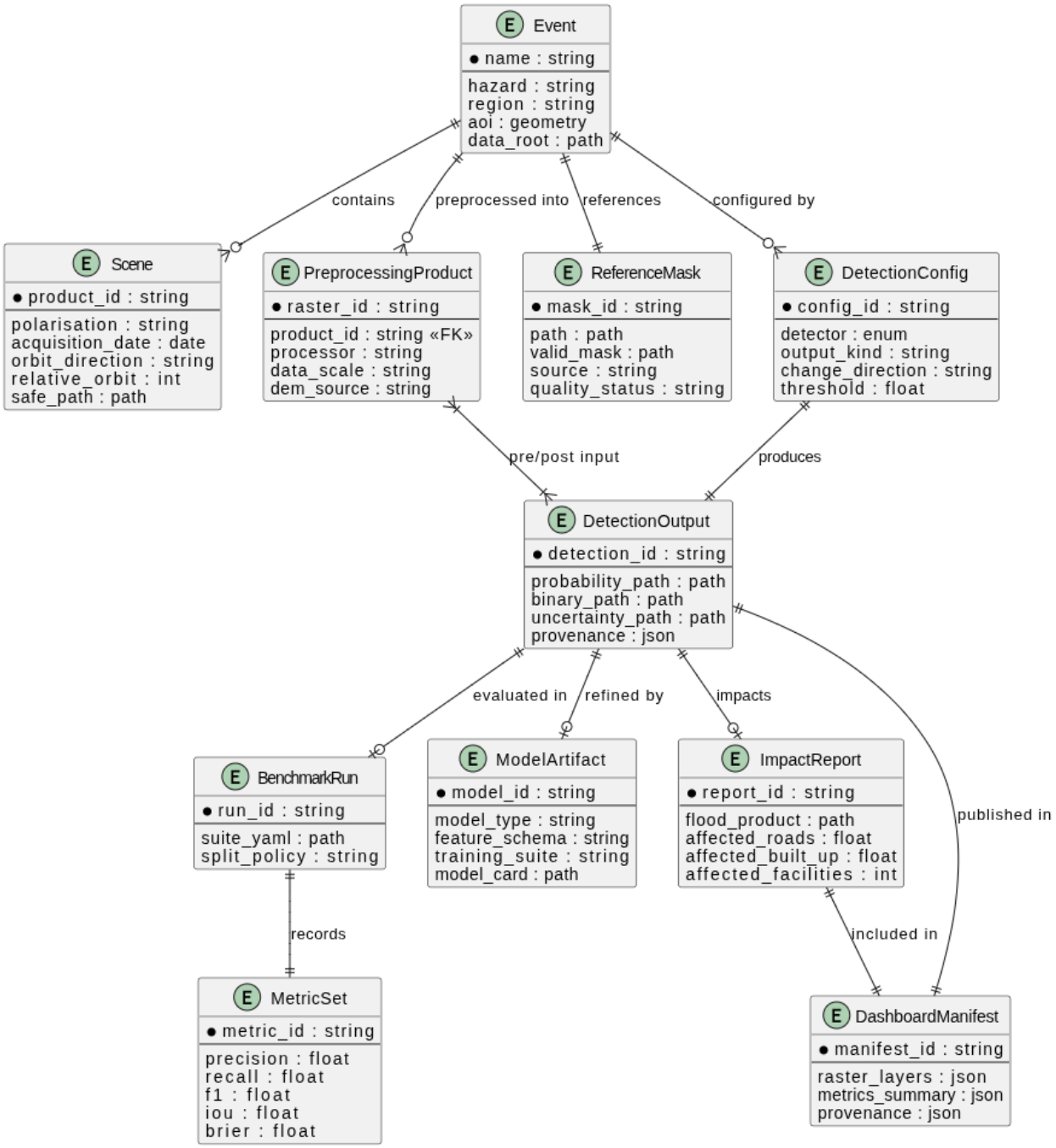


Figure 4.2. Entity-relationship diagram

Class Diagram

The class diagram shows important configuration and result classes including EventWorkflowConfig, EventEvaluationConfig, DetectionConfig, AdvancedFloodConfig, AdvancedFloodResult, RefinementConfig, BenchmarkSuite, and ImpactReport.

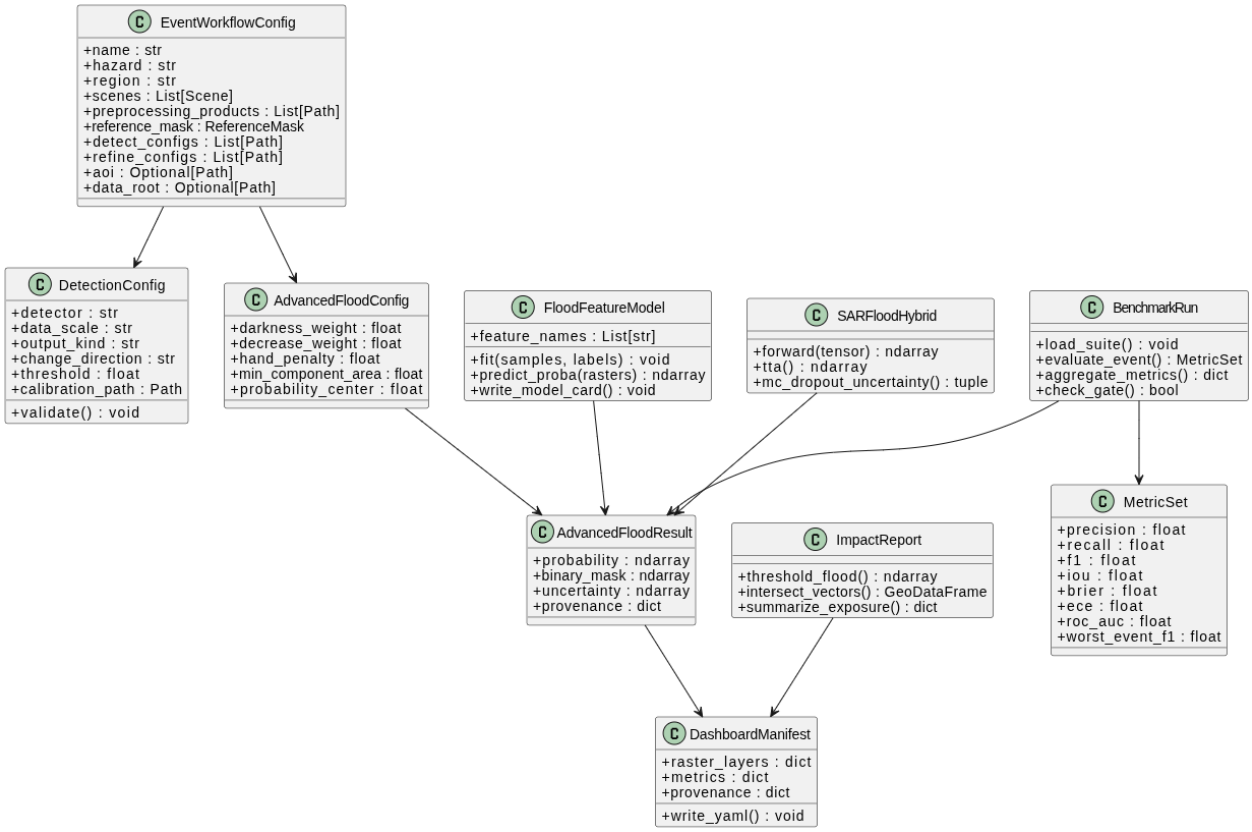


Figure 4.3. Class diagram

Sequence Diagrams

Separate sequence diagrams are used for the major implemented workflows: flood detection, feature-model promotion, benchmark-suite execution, and impact/dashboard review. This keeps data ownership, artifacts, and review boundaries clear.

Figure 4.4.1 — Sequence Diagram: Flood Detection (Improved Lifecycle)

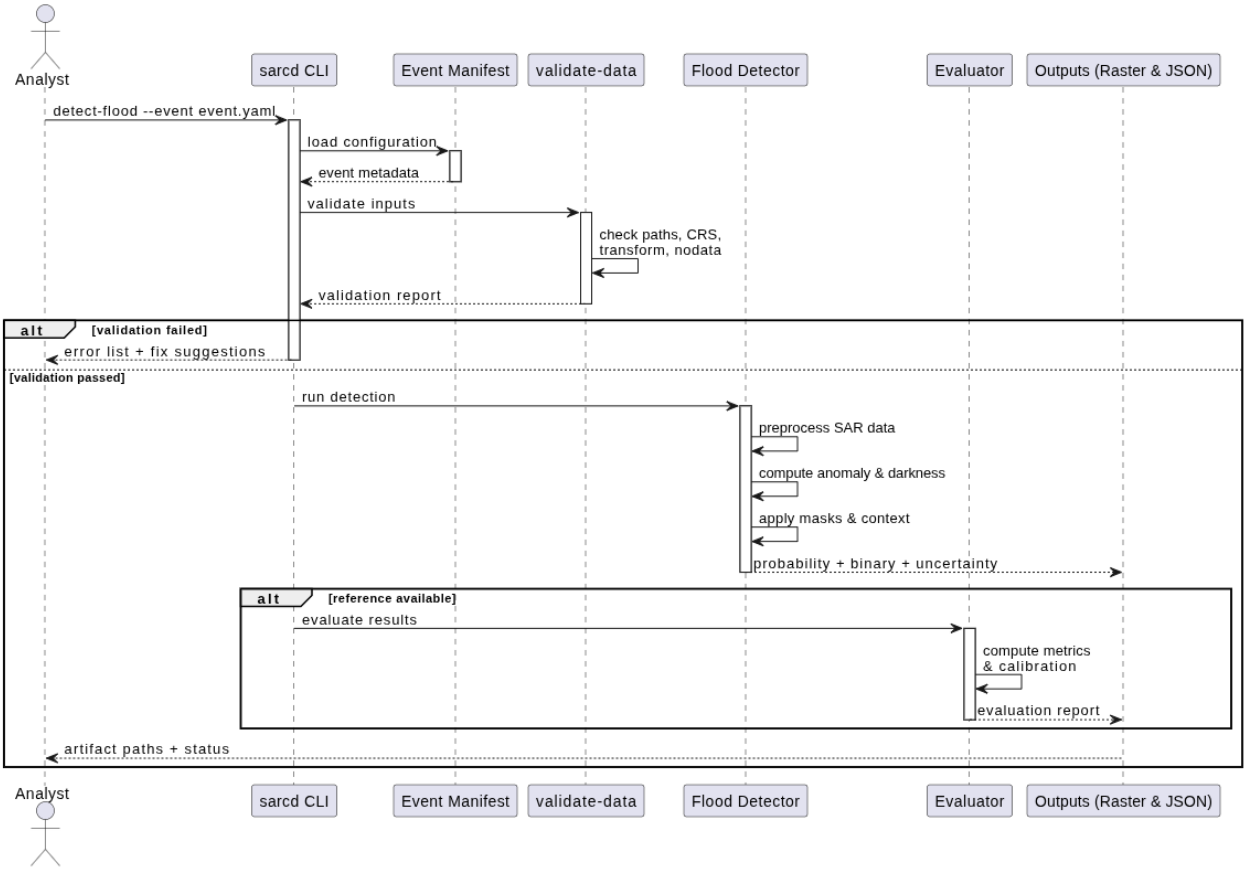


Figure 4.4.1. Sequence diagram for flood detection

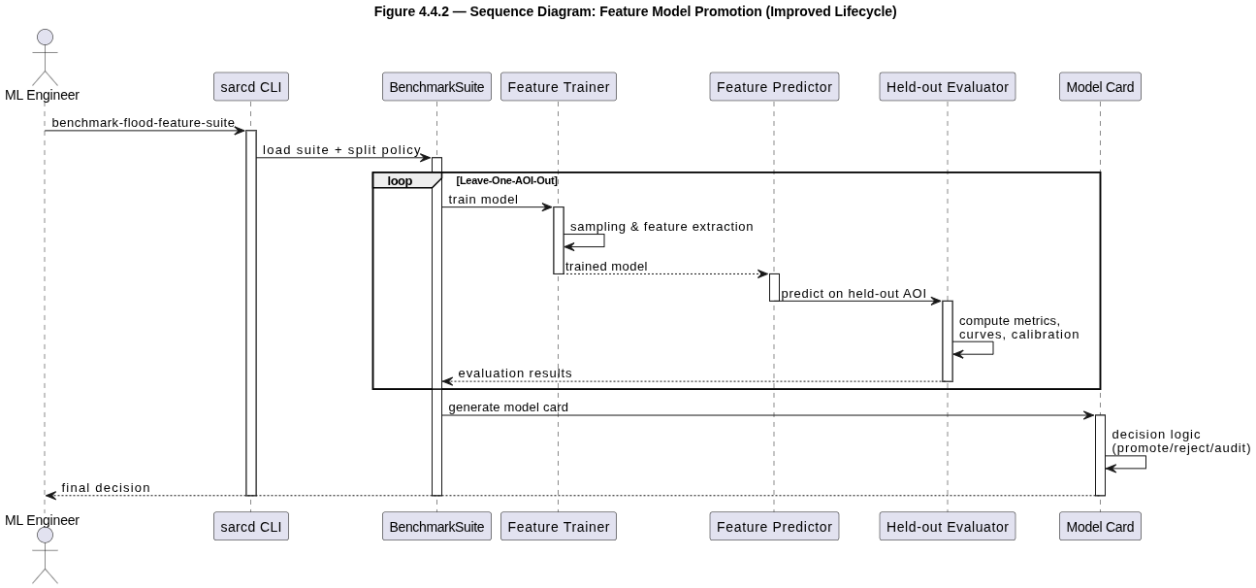


Figure 4.4.2. Sequence diagram for feature-model promotion

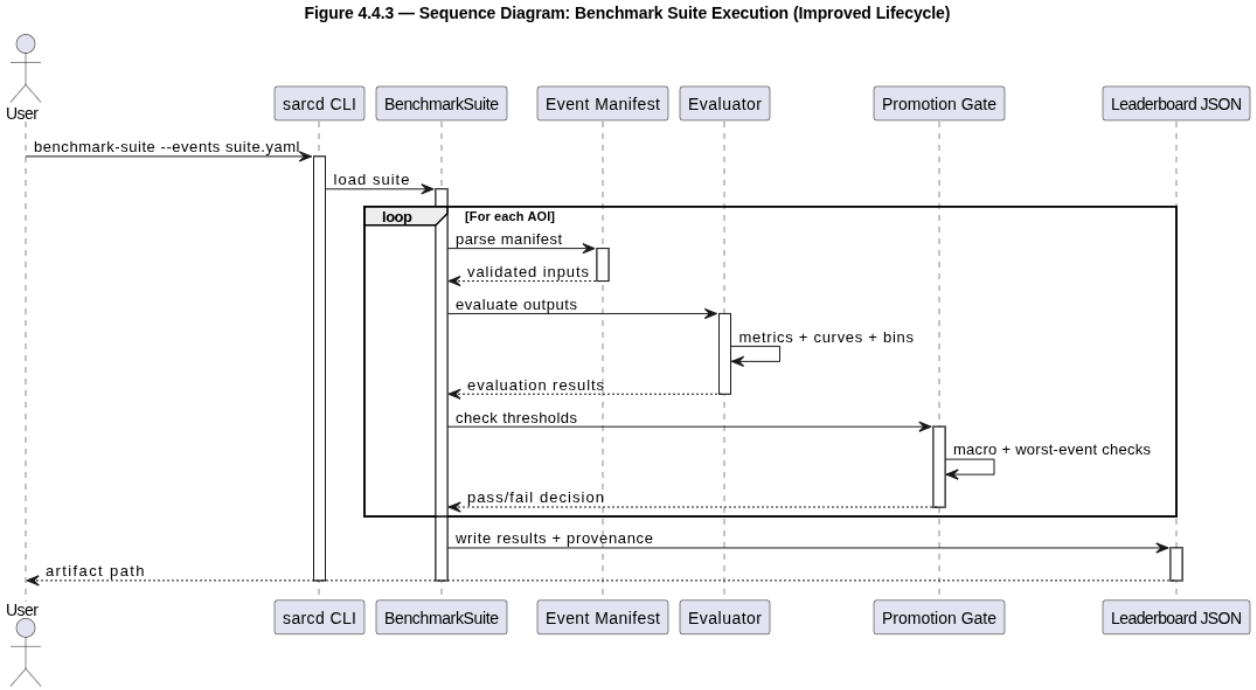


Figure 4.4.3. Sequence diagram for benchmark-suite execution

Figure 4.4.4 — Sequence Diagram: Impact & Dashboard Review (Improved Lifecycle)

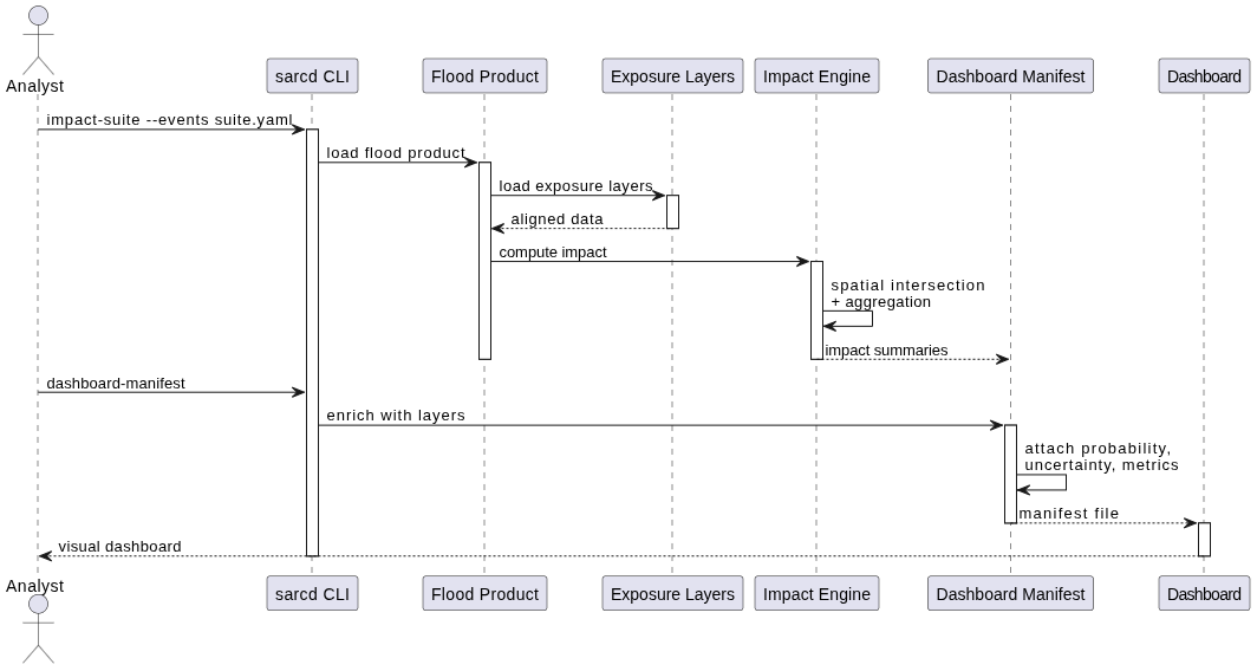


Figure 4.4.4. Sequence diagram for impact and dashboard review

Activity Diagrams

The activity diagram shows the decision flow: validate event, check data availability, choose production RTC/SNAP path, run detector, evaluate if reference exists, calibrate if continuous output, write provenance, and publish dashboard manifest.

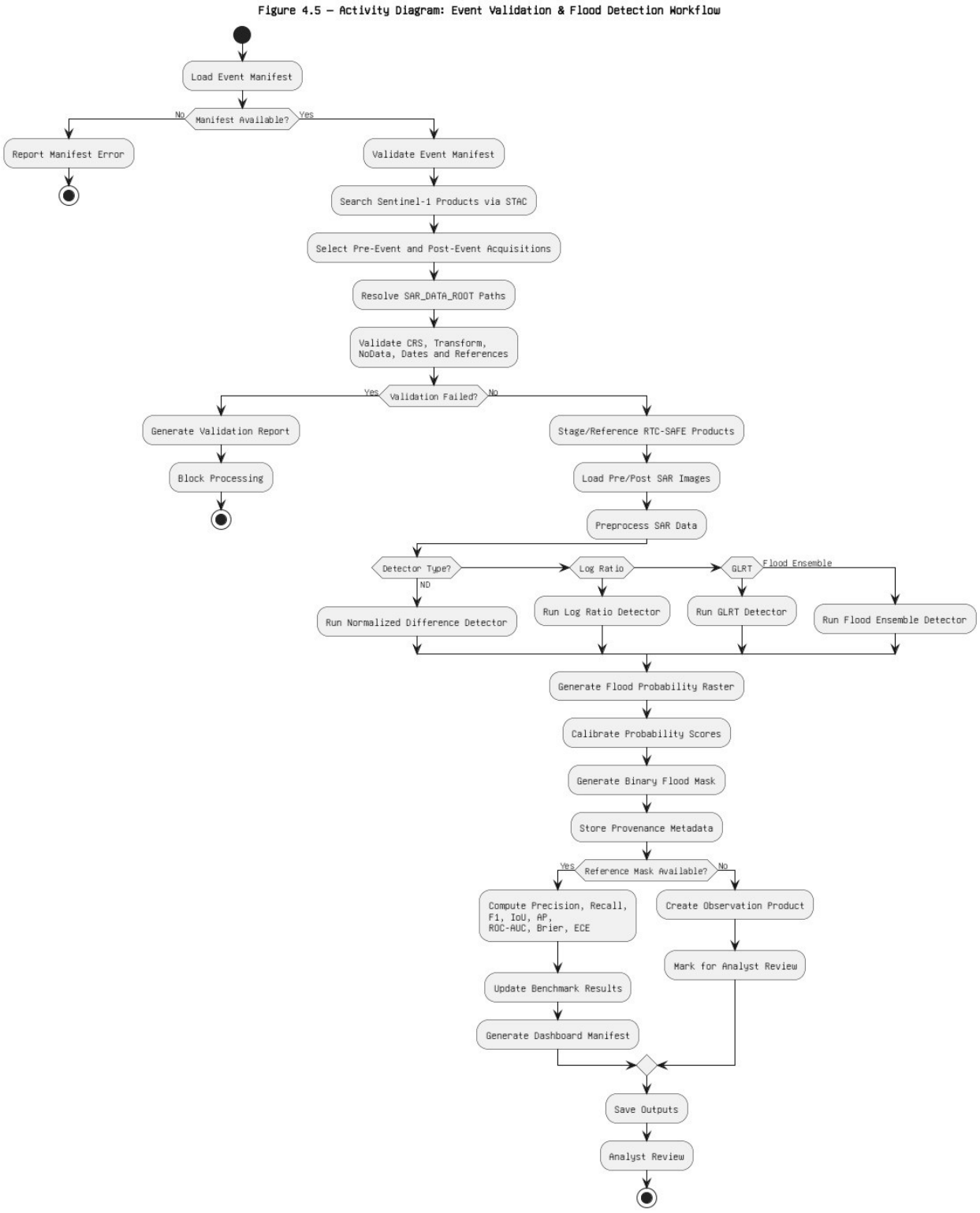


Figure 4.5. Activity diagram for event validation and detection

Figure 4.5.1 - Activity Diagram: Flood Detection Workflow

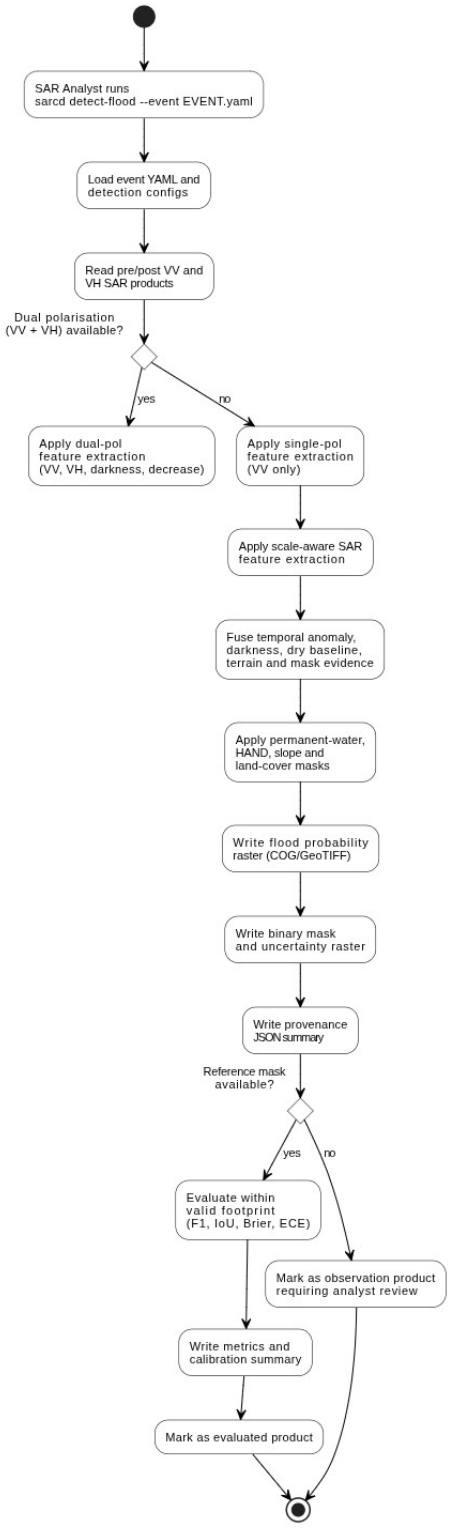


Figure 4.5.1. Activity diagram for flood detection workflow

Figure 4.5.2 - Activity Diagram: ML Refinement and Model Promotion

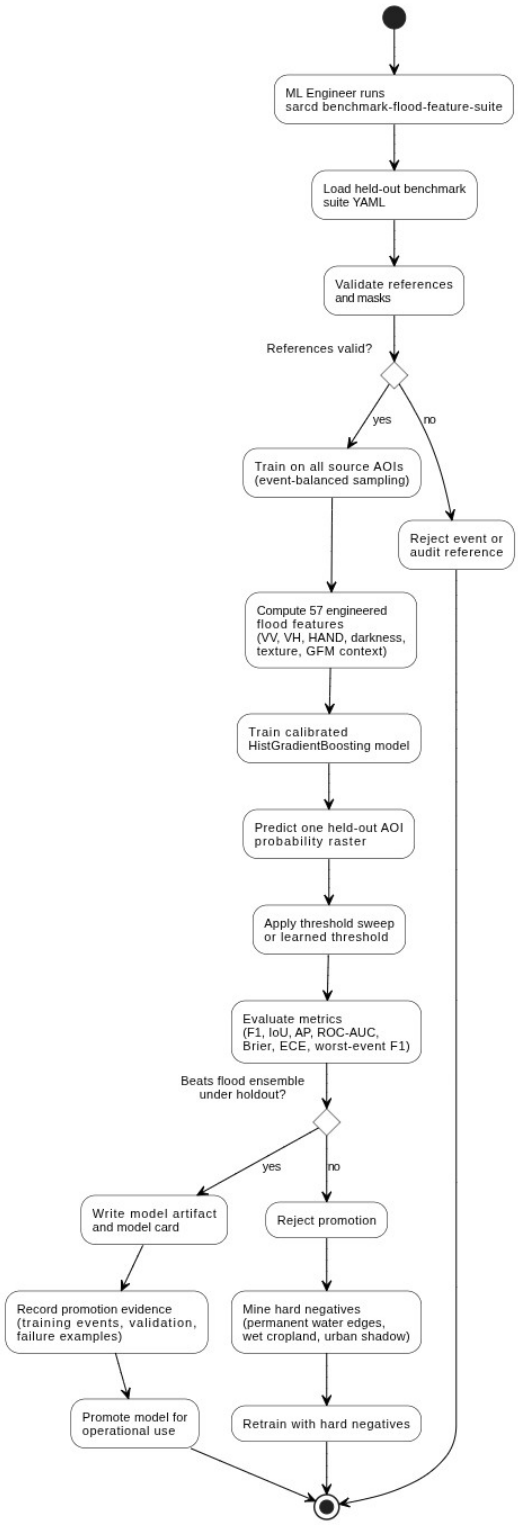


Figure 4.5.2. Activity diagram for ML refinement and model promotion

Figure 4.5.3 - Activity Diagram: Benchmark Suite Execution

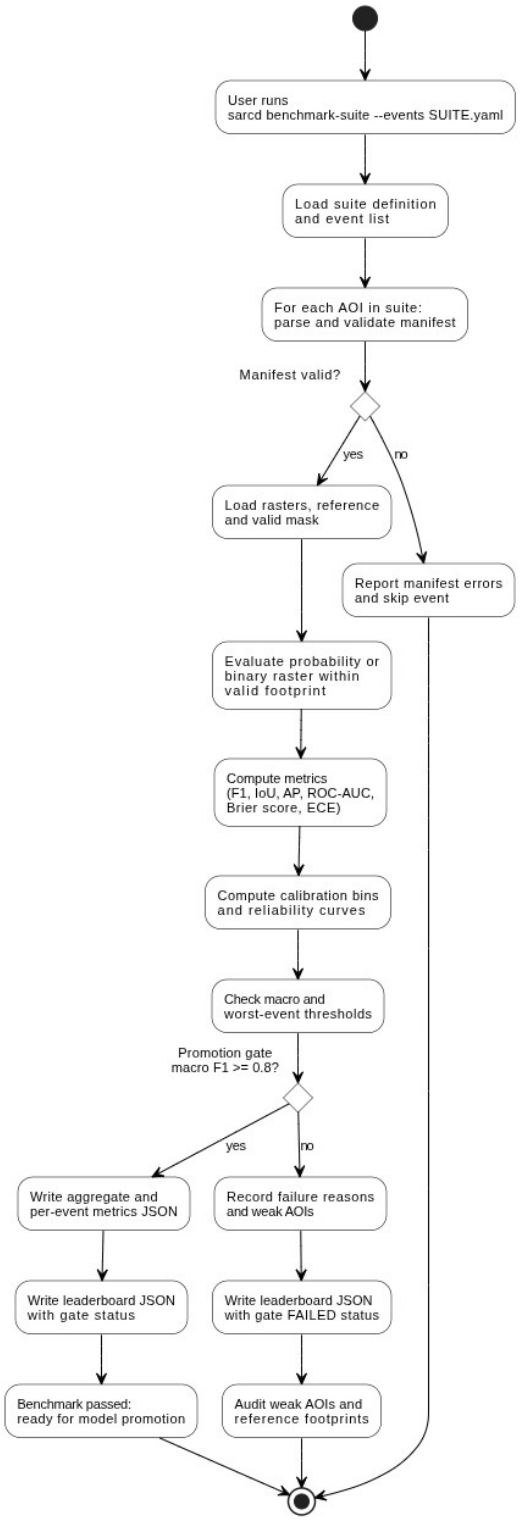


Figure 4.5.3. Activity diagram for benchmark-suite execution

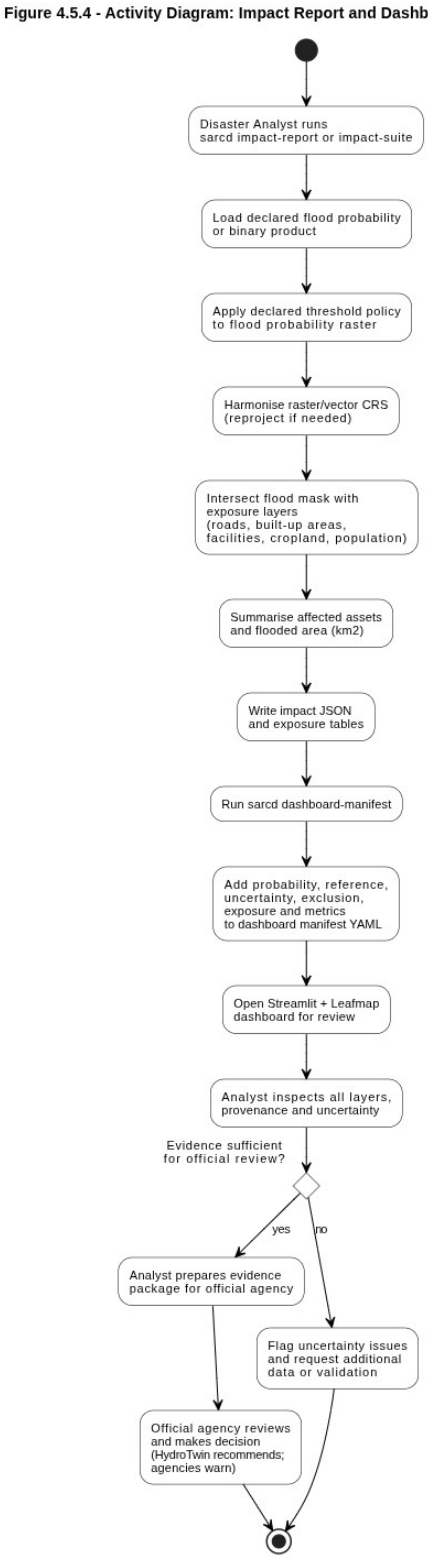


Figure 4.5.4. Activity diagram for impact report and dashboard review

Data Flow Diagrams

Figures 4.6 and 4.7 document DFD Level 0 and Level 1. The Level-2 diagrams are split into the three implemented areas where data-flow detail adds value: flood detection, benchmark/model promotion, and impact/dashboard review. This avoids adding diagrams for workflows that are only conceptual.

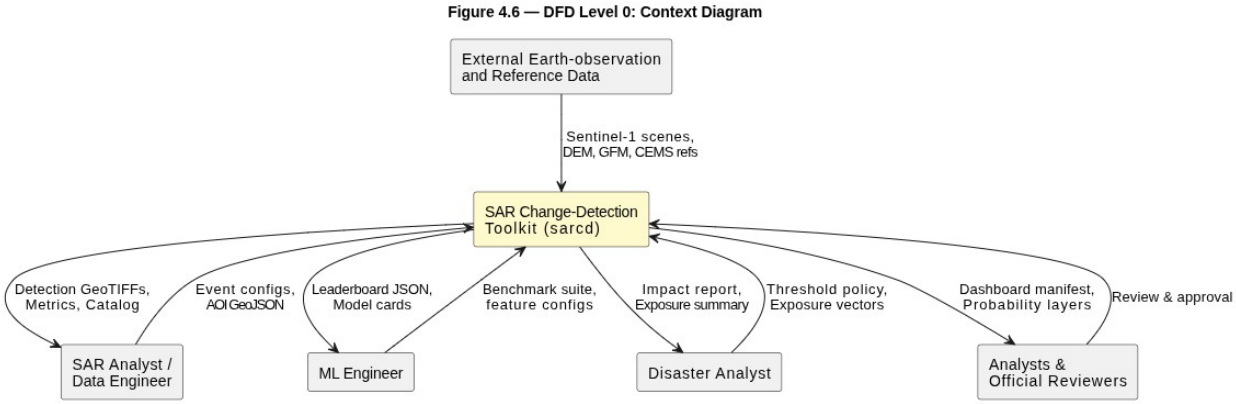


Figure 4.6. DFD Level 0

Figure 4.7 — DFD Level 1: System Decomposition

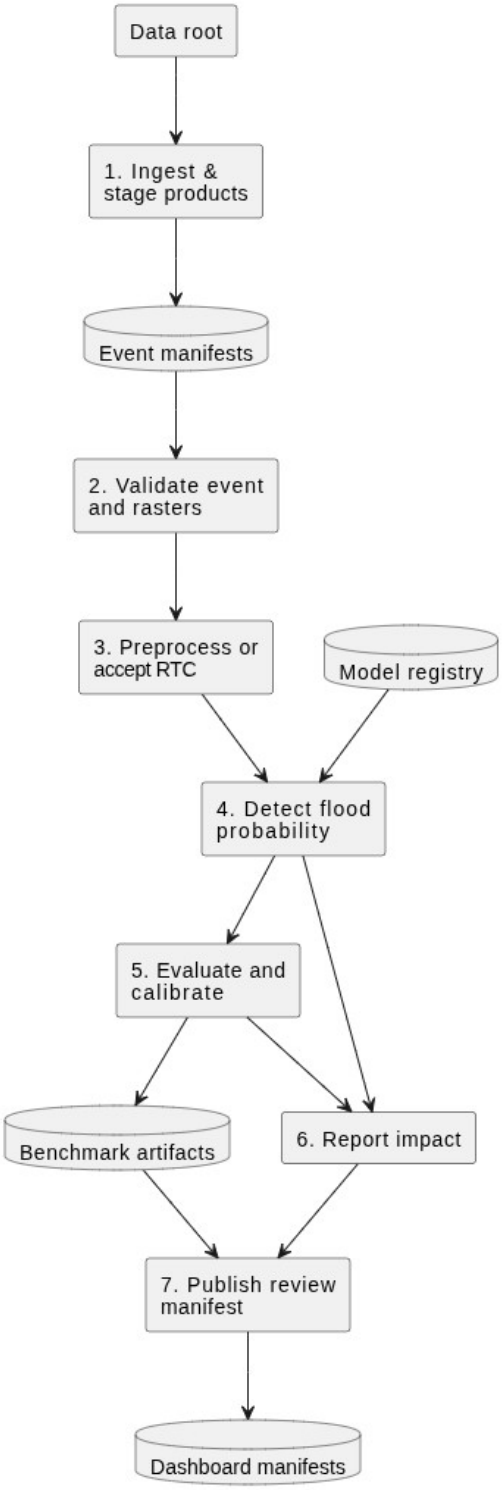


Figure 4.7. DFD Level 1

Figure 4.8.1 — DFD Level 2: Flood Detection Process

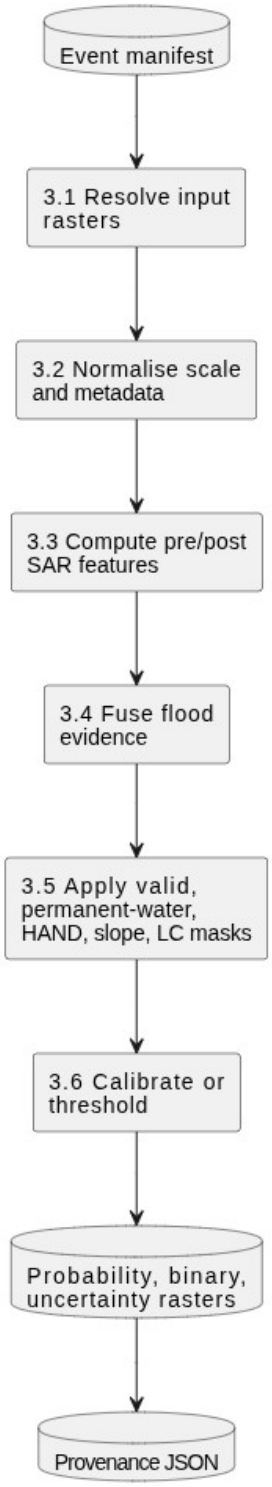


Figure 4.8.1. DFD Level 2 for flood detection

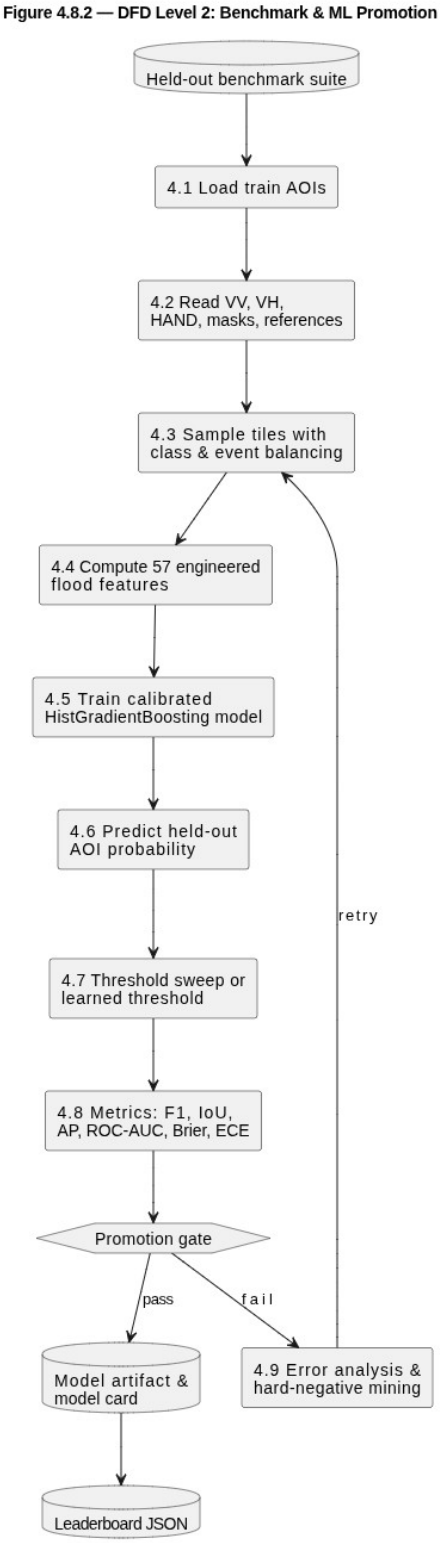


Figure 4.8.2. DFD Level 2 for benchmark and ML promotion

Figure 4.8.3 — DFD Level 2: Impact & Dashboard Review

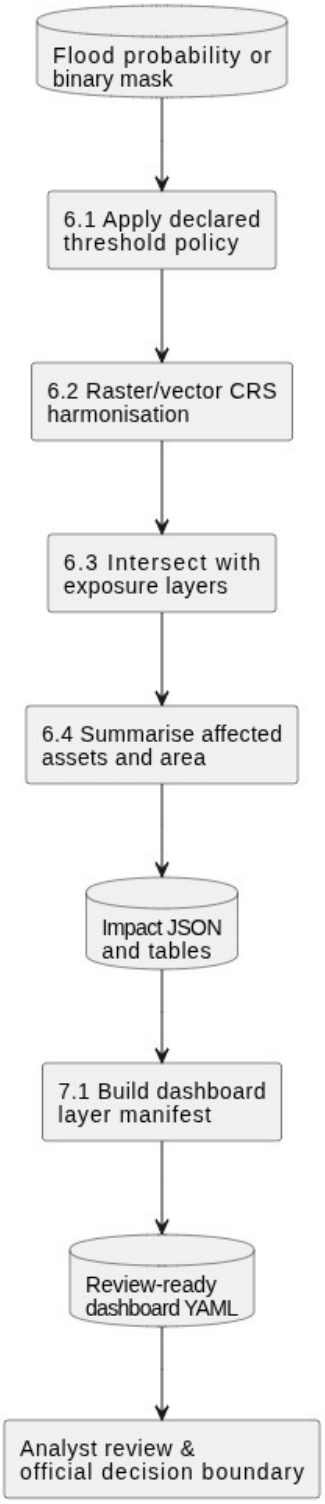


Figure 4.8.3. DFD Level 2 for impact and dashboard review

Flood Detector Semantics

The detection configuration explicitly defines hazard, detector, data scale, change direction, output kind, threshold, and calibration path. The supported detector names include log_ratio, coh_change, glrt, normalized_difference, and flood_ensemble. For flood use, the expected SAR change direction is usually a backscatter decrease in open water or wet smooth surfaces, with exceptions in urban or vegetated areas.

Table 6.1. Detector Semantics and Scientific Interpretation

Field	Allowed or Expected Values	Scientific Role
hazard	flood, landslide, generic	Prevents flood-specific thresholds from being applied to unrelated hazards
detector	log_ratio, normalized_difference, glrt, flood_ensemble	Selects transparent physics baseline or fused flood model
data_scale	linear_power, db	Avoids invalid arithmetic across log and linear backscatter
change_direction	decrease, increase, both	Encodes whether the event is expected to darken or brighten SAR response
output_kind	score, probability, binary	Controls whether evaluation needs a threshold or calibration file
threshold/calibration_path	numeric or JSON path	Makes conversion from score/probability to binary mask reproducible

Advanced Flood Probability Model

The advanced flood detector combines VV and VH evidence where available. Core features include post-event darkness, pre-to-post decrease, normalized decrease, cross-polarization behavior, optional HAND penalties, permanent-water exclusion, connected-component cleanup, and probability calibration. The output is a probability or binary flood mask depending on configuration.

The intended production chain is conservative. First, pre-event and post-event SAR products are accepted only when they are RTC/ARD or produced by a SNAP graph with orbit correction, calibration, speckle filtering, and terrain correction. Second, pixel scores are interpreted with masks for permanent water, steep terrain, invalid reference regions, and low-confidence ancillary zones. Third, continuous outputs are calibrated before being used as probabilities. Finally, component filtering and area summaries are written as postprocessing metadata so small artifacts are recorded before any operational flood polygon review.

Feature-Based ML Classifier

The flood feature model uses a structured feature schema rather than a black-box-only input. Features include VV/VH dB values, decreases and increases, normalized decrease, darkness metrics, cross-ratio metrics, dual-pol agreement, HAND, local mean context windows, local texture, and GFM probability/context variables. Training supports event-class-balanced sampling so one large AOI does not dominate the model.

Table 6.2. Flood Feature Families

Feature Family	Examples	Failure Mode Controlled
Backscatter state	pre/post VV, pre/post VH, dB darkness	Separates water-like post-event surfaces from ordinary land
Temporal anomaly	VV/VH decrease, increase, normalized decrease	Detects flood-induced change instead of static low backscatter only
Dual-pol relationship	cross-ratio, agreement, inverse-variance fusion	Reduces false positives from single-polarization artifacts
Terrain context	HAND, slope, local elevation proxy	Suppresses implausible high-terrain flood detections
Spatial context	5, 11, and 21 pixel local means and texture	Uses neighborhood consistency and filters isolated

		speckle
Reference/product context	GFM probability, extent, valid and local GFM context	Uses high-confidence external evidence where available

Deep Learning Refinement

The deep-learning module standardizes on a SAR flood hybrid architecture. It uses convolutional residual blocks, squeeze-excite attention, atrous spatial pyramid pooling, attention-gated skip connections, dropout, and Monte Carlo dropout at inference time for uncertainty. The 16 GB GPU target is suitable for mixed-precision 256 to 512 pixel tiles with gradient accumulation.

Mathematical Formulation

- Log ratio score: L = log(post + eps) - log(pre + eps), interpreted with detector-specific direction.
- Normalized difference: ND = (post - pre) / (post + pre + eps), with direction-aware thresholding.
- Probability calibration: p_cal = f(p_raw), where f is fitted on validation outputs and reference labels.
- F1 = 2 * precision * recall / (precision + recall), used as the primary benchmark gate.
- IoU = TP / (TP + FP + FN), also reported as critical success index for flood extent overlap.
- Brier score = mean((p - y)^2), used to assess probability quality.

Chapter 5: Implementation

Development Environment

The repository is a Python package installed locally through uv or a virtual environment. The pyproject file declares Python 3.10+, core geospatial/scientific dependencies, optional ML and forecasting extras, the sarcd command-line entry point, Ruff lint settings, and pytest settings.

Technology Stack

Table 7. Technology Stack

Layer	Technology	Purpose
CLI	Typer, Rich	Structured commands and readable terminal output
Raster processing	Rasterio, NumPy, SciPy	Read, write, transform, and evaluate rasters
Configuration	PyYAML, dataclasses	Event, detector, benchmark, and ML configs
Ingestion	Requests, pystac-client	STAC/OData/RTC product discovery and staging
Machine learning	scikit-learn, PyTorch optional	Feature classifier and neural refinement
Dashboard	Streamlit, Leafmap	Local map review and layer controls
Testing	pytest, Ruff	Unit/integration quality checks
Outputs	GeoTIFF/COG, JSON, YAML	Rasters, provenance, metrics, dashboard manifests

Module Description

Table 8. Implementation Modules

Module	Responsibility	Key Files
ingest	STAC/OData/GFM/SAFE/Planetary Computer staging	ingest/stac.py, odata.py, gfm.py, safe.py, planetary_computer.py
preprocess	SNAP GPT graph building and Python preprocessing operations	preprocess/snap_processor.py, pipeline.py, operations.py
events	Event manifest schema and validation checks	events/config.py, validation.py
detect	Detector configs, flood algorithms, postprocessing, feature classifier	detect/config.py, flood_advanced.py, flood_features.py, pipeline.py
ml	Neural SAR flood refinement, augmentation, training	ml/deep_learning.py, training.py, pipeline.py
benchmark	Event and raster suite evaluation, gates, leaderboards	benchmark.py
impact	Flood/exposure intersection and summary reports	impact.py
dashboard	Manifest-driven local visualization	dashboard/app.py, config.py, runtime.py
forecasting	Experimental risk features, conformal calibration, models	forecasting/*.py
utils	Evaluation, IO, catalog, manifest, landcover, error analysis	utils/*.py

Implementation Coverage Audit

A full repository pass shows that the implemented project is broader than one detector script. The strongest and most validated path is the Pakistan flood workflow; forecasting, fusion, and severity modules are present but remain experimental compared with the flood benchmark core.

Table 8.1. Implementation Coverage Audit

Capability	Implemented Evidence	Maturity
Runtime checks	sarcd doctor checks Python, rasterio, SNAP GPT, data root, and Copernicus credentials	Implemented and tested
Data validation	validate-data and events/validation.py check manifests, raster paths, georeferencing, absolute paths, scenes, and reference quality	Implemented and tested

CDSE access	ingest/stac.py and ingest/odata.py implement STAC search, OAuth token handling, OData URL construction, and download errors	Implemented; network/data availability dependent
GFM/CEMS staging	ingest/gfm.py and reference.py stage GFM/CEMS masks, valid footprints, likelihood/probability layers, and mask algebra	Implemented and covered by tests
SNAP preprocessing	snap_processor.py builds ESA SNAP GPT graphs for orbit, calibration, speckle filtering, terrain correction, and GeoTIFF writing	Implemented; requires local SNAP GPT
Flood detection	detect/config.py, pipeline.py, flood_advanced.py, and postprocess.py implement flood semantics, masking, dual-pol probability, overviews, and summaries	Production path for this project
Feature ML	detect/flood_features.py trains event-level feature models and writes probability rasters	Validated on Pakistan benchmark artifacts
Deep learning	ml/deep_learning.py and ml/pipeline.py implement SAR flood hybrid network, tiling, mixed precision, losses, and uncertainty	Implemented; refinement stage, not primary gate
Benchmarking	benchmark.py supports event/raster suites, transfer tests, learned thresholds, gates, object metrics, calibration, and reference quality	Core scientific validation path
Impact reporting	impact.py intersects flood rasters with CEMS exposure layers and reports affected features/areas	Implemented for CEMS-style vectors
Dashboard	dashboard/runtime.py uses downsampled previews and streaming threshold masks instead of loading full rasters blindly	Implemented local review interface
Forecasting/fusion/severity	forecasting, fusion, and severity packages exist with tests marked as experimental specs where APIs remain in design	Experimental research modules

Code Evidence Screenshots

The following screenshots are included as implementation evidence. They are not decorative diagrams; they show the exact source modules that define the command surface, detector semantics, flood probability logic, feature schema, benchmark gate policy, event manifest schema, SNAP graph generation, and dashboard raster preview behavior.

CLI command surface: doctor, validation, detection, benchmark

```
712  @app.command("doctor")
713  def doctor(
714      strict: bool = typer.Option(
715          False,
716          "--strict/--no-strict",
717          help="Exit with an error when required production dependencies are missing.",
718      ),
719      online: bool = typer.Option(
720          False,
721          "--online/--offline",
722          help="Also verify live CDSE STAC reachability and OAuth token exchange.",
723      ),
724  ) → None:
725      """Inspect local runtime dependencies and data-access settings."""
 ⋯   # middle lines omitted for report legibility
2762          help="Overwrite existing detection outputs.",
 ⋯
2894  @app.command("benchmark-suite")
2895  def benchmark_suite_command(
2896      events: Optional[List[Path]] = typer.Option(
2897          None,
2898          "--event",
2899          "-e",
2900          exists=True,
2901          help="Event manifest YAML. Repeat for a multi-event Pakistan suite.",
2902      ),
2903      manifest: Optional[Path] = typer.Option(
2904          None,
```

Figure 5.1. CLI runtime checks and data-validation commands

Detector semantics: hazard, scale, direction, output kind

```
66  class DetectionConfig:
67      output_dir: Path
68      pre_event: Optional[Path] = None
69      post_event: Optional[Path] = None
70      pre_event_vh: Optional[Path] = None
71      post_event_vh: Optional[Path] = None
72      detector: str = "log_ratio"
73      hazard: str = "generic"
74      data_scale: str = "linear_power"
75      change_direction: str = "both"
76      output_kind: str = "score"
77      eps: float = 1e-6
78      looks: float = 1.0
79      mask: Optional[Path] = None
...   # middle lines omitted for report legibility
158             raise ValueError(f"output_kind must be one of {sorted(VALID_OUTPUT_KINDS)}.")
159         if self.detector == "flood_ensemble" and self.hazard != "flood":
160             self.hazard = "flood"
161         if self.hazard == "flood" and self.change_direction == "both":
162             self.change_direction = "decrease"
163         if (
164             self.threshold is not None
165             and self.threshold < 0
166             and self.output_kind != "binary"
167         ):
168             raise ValueError(
169                 "Negative thresholds are ambiguous for continuous detector outputs. "
170                 "Use change_direction='decrease' with a positive magnitude threshold."
```

Figure 5.2. Detector semantics and validation domains

Advanced flood probability: dual-pol evidence, darkness, masks

```
135 def advanced_flood_probability(
136     pre_vv: np.ndarray,
137     post_vv: np.ndarray,
138     *,
139     pre_vh: Optional[np.ndarray] = None,
140     post_vh: Optional[np.ndarray] = None,
141     hand: Optional[np.ndarray] = None,
142     config: Optional[AdvancedFloodConfig] = None,
143 ) -> AdvancedFloodResult:
144     """Compute advanced flood score, probability, and confidence rasters."""
145     cfg = config or AdvancedFloodConfig()
146     vv_change = directional_decrease(
147         pre_vv,
148         post_vv,
...   # middle lines omitted for report legibility
201         denom = vv_change + vh_change + cfg.eps
202         agreement[valid] = 1.0 - np.abs(vv_change[valid] - vh_change[valid]) / denom[valid]
203         confidence = probability * np.clip(0.5 + 0.5 * agreement, 0.0, 1.0)
204     else:
205         confidence = probability.copy()
206
207     metadata.update(cfg.metadata)
208     metadata["score_kind"] = "dual_pol_darkness_hand_flood_likelihood"
209     metadata["probability_center"] = float(cfg.probability_center)
210     metadata["probability_scale"] = float(cfg.probability_scale)
211
212     return AdvancedFloodResult(
213         probability=probability.astype(np.float32, copy=False),
```

Figure 5.3. Advanced flood probability from SAR evidence

Feature-based flood model: interpretable SAR/context features

```
FLOOD_FEATURE_NAMES = [
    "pre_vv_db",
    "post_vv_db",
    "vv_decrease_db",
    "vv_increase_db",
    "vv_abs_change_db",
    "vv_norm_decrease",
    "post_vv_darkness_db",
    "pre_vv_darkness_db",
    "pre_vh_db",
    "post_vh_db",
    "vh_decrease_db",
    "vh_increase_db",
    "vh_abs_change_db",
... # middle lines omitted for report legibility
    "dual_decrease_mean_5",
    "post_vv_darkness_mean_5",
    "post_vh_darkness_mean_5",
    "vv_decrease_local_contrast_5",
    "dual_decrease_support_5",
    "post_vv_dark_support_5",
    "gfm_probability",
    "gfm_extent",
    "gfm_valid",
    "gfm_probability_mean_5",
    "gfm_extent_mean_5",
    "vv_decrease_mean_11",
    "vh_decrease_mean_11",
```

Figure 5.4. Feature-based flood classifier input schema

Benchmark gates: reference-quality policy and worst-event control

```
class BenchmarkGate:
    """Minimum benchmark requirements for a suite run."""

    min_f1: Optional[float] = None
    min_iou: Optional[float] = None
    min_average_precision: Optional[float] = None
    min_events: int = 1
    reference_quality: str = "trusted"

    def __post_init__(self) -> None:
        self.reference_quality = _normalize_reference_quality_policy(self.reference_quality)

    @property
    def thresholds(self) -> Dict[str, float]:
... # middle lines omitted for report legibility
            "eligible_events": eligible_events,
            "excluded_events": excluded_events,
            "n_eligible_entries": len(eligible_entries),
...
        "passed": len(events_passing) ≥ gate.min_events,
        "thresholds": thresholds,
        "min_events": gate.min_events,
        "reference_quality": reference_policy,
        "eligible_events": eligible_events,
        "excluded_events": excluded_events,
        "n_eligible_entries": len(eligible_entries),
        "n_excluded_entries": len(excluded_entries),
        "events_passing": events_passing,
```

Figure 5.5. Benchmark gates and reference-quality policy

Event manifest contract: paths, scenes, products, references

```
class EventWorkflowConfig:
    name: str
    description: Optional[str]
    preprocess_config: Optional[Path]
    detect_configs: List[Path]
    evaluation: Optional[EventEvaluationConfig]
    refine_configs: List[Path] = field(default_factory=list)
    hazard: str = "generic"
    region: Optional[str] = None
    split: str = "benchmark"
    tags: List[str] = field(default_factory=list)
    aoi: Optional[Path] = None
    data_root: Optional[Path] = None
    source_path: Optional[Path] = None
... # middle lines omitted for report legibility
            reference_raw = eval_section.get("reference")
            if not reference_raw:
                raise ValueError("Evaluation config requires a 'reference' path.")
            positive_label = float(eval_section.get("positive_label", 1.0))
            threshold = eval_section.get("threshold")
            output_dir_raw = eval_section.get("output_dir")
            valid_mask_raw = eval_section.get("valid_mask")
            evaluation_cfg = EventEvaluationConfig(
                reference=cls._resolve_path(config_path, reference_raw, data_root=data_root),
                positive_label=positive_label,
                threshold=float(threshold) if threshold is not None else None,
                output_dir=(
                    cls._resolve_path(config_path, output_dir_raw, data_root=data_root)
```

Figure 5.6. Event manifest dataclasses for reproducible workflows

SNAP GPT preprocessing graph: orbit, calibration, speckle, terrain

```
def create_grd_preprocessing_graph(config: SnapConfig, output_path: Path) → str:
    """Create SNAP XML graph for GRD preprocessing.

    Processing chain:
    1. Read SAFE product
    2. Apply-Orbit-File (precise orbits)
    3. Remove-GRD-Border-Noise
    4. Calibration (to Sigma0)
    5. Speckle-Filter
    6. Terrain-Correction (with DEM)
    7. Write to GeoTIFF
...
    # 2. Apply-Orbit-File
    if config.apply_orbit_file:
... # middle lines omitted for report legibility
        "enl": "1.0",
        "numLooksStr": "1",
        "windowSize": config.speckle_window_size,
        "targetWindowSizeStr": "3x3",
        "sigmaStr": "0.9",
        "anSize": "50"
    }, last_node)

    # 6. Terrain Correction
    last_node = add_node("Terrain-Correction", {
        "demName": config.dem_name,
        "externalDEMNoDataValue": "0.0",
        "externalDEMApplyEGM": "true",
```

Figure 5.7. SNAP GPT preprocessing graph construction

Dashboard previews and masks: downsampled reads and block windows

```
def _read_array_and_profile(path_str: str) → Tuple[np.ndarray, dict]:
    path = Path(path_str)
    with rasterio.open(path) as src:
        scale = max(1.0, np.sqrt((src.width * src.height) / MAX_PREVIEW_PIXELS))
        out_width = max(1, int(src.width / scale))
        out_height = max(1, int(src.height / scale))
        data = src.read(
            1,
            out_shape=(out_height, out_width),
            resampling=Resampling.average,
        ).astype(np.float32)
        nodata = src.nodata
        if nodata is not None:
            data = np.where(np.isclose(data, nodata), np.nan, data)
... # middle lines omitted for report legibility
...
        with MemoryFile() as memfile:
            with memfile.open(**profile) as dst:
                for _, window in src.block_windows(1):
                    data = src.read(1, window=window).astype(np.float32)
                    nodata = src.nodata
                    if nodata is not None:
                        data = np.where(np.isclose(data, nodata), np.nan, data)
                    mask = np.full(data.shape, nodata_value, dtype=np.uint8)
                    finite = np.isfinite(data)
                    mask[finite] = (data[finite] ⩾ threshold).astype(np.uint8)
                    dst.write(mask, 1, window=window)
            payload = memfile.read()
```

Figure 5.8. Dashboard raster preview downsampling

User Interface and Implementation Evidence

The primary user interface is the sarcd command-line application. It exposes environment checks, data validation, product staging, flood detection, benchmarking, model training, calibration, impact reporting, and dashboard-manifest generation as reproducible commands. The secondary interface is the local dashboard, which consumes generated manifests and supports analyst review of probability, binary mask, reference, uncertainty, exclusion, exposure, metrics, and provenance layers.

Command-Line Interface

Table 9. Important sarcd Commands

Command	Purpose
sarcd doctor	Check local environment, dependencies, credentials, SNAP GPT, and data root assumptions
sarcd validate-data --event EVENT.yaml	Validate event manifest, paths, alignment, CRS, nodata, and metadata
sarcd detect-flood --event EVENT.yaml	Run flood detector from event manifest
sarcd benchmark-suite --events SUITE.yaml	Run multi-event benchmark and produce leaderboard JSON
sarcd train-flood-feature-model	Train feature-based flood classifier
sarcd apply-flood-feature-model	Apply trained flood feature classifier to AOIs
sarcd fit-probability-calibration	Fit calibration mapping for probability rasters
sarcd impact-report	Intersect flood products with exposure layers
sarcd dashboard-manifest	Generate dashboard-ready manifest from outputs

Output Standardization

- Raster outputs are GeoTIFF/COG products where possible.
- Sidecar JSON summaries record thresholds, metrics, provenance, calibration, and reference quality.
- Benchmark outputs use machine-readable JSON with aggregate and per-event metrics.

- Dashboard manifests are created from actual outputs rather than manually typed metrics.
- Event manifests can resolve data under SAR_DATA_ROOT to avoid hardcoding local disk paths.

Chapter 6: Testing and Validation

Testing is divided into unit tests, integration tests, quality checks, and real-data benchmark validation. The codebase includes tests for detection configs and pipelines, evaluation metrics, event manifests, GFM ingestion, impact reporting, probability calibration, preprocessing, dashboard configs, ML refinement, and absolute-path hygiene.

Unit Testing

Table 10. Unit Test Coverage Summary

Area	Representative Tests	Risk Controlled
Detection	test_detection_config.py, test_detection_pipeline.py, test_flood_features.py	Detector routing, thresholds, flood feature correctness
Evaluation	test_evaluate.py, test_probability_calibration.py	Metric correctness and probability calibration
Manifests	test_manifest.py, test_no_absolute_paths.py	Portable configs and manifest semantics
Ingestion	test_ingest_stac.py, test_ingest_odata.py, test_gfm_ingest.py	Data access request logic and reference staging
Preprocessing	test_preprocess_config.py, test_preprocess_pipeline.py	Raster operations and provenance flow
ML	test_deep_learning.py, test_ml_refinement.py	Model construction, tiling, inference, and refinement outputs
Impact	test_impact.py	Exposure intersection and summary logic

Integration Testing

Table 11. Integration Test Cases

ID	Scenario	Expected Result
IT-01	Run event configuration through validation and detection	Outputs are produced or missing inputs are reported with actionable errors
IT-02	Evaluate continuous raster without threshold	System rejects ambiguous evaluation unless calibration or threshold is supplied
IT-03	Run benchmark suite across multiple event manifests	Leaderboard JSON includes aggregate, per-event, and gate sections
IT-04	Generate dashboard manifest from catalog outputs	Manifest references real rasters and summaries
IT-05	Run impact suite with vector exposure layers	Affected feature counts and flooded areas are reported

Quality Testing

- Ruff is scoped to meaningful syntax/undefined-name rules in src and tests.
- No-absolute-path tests protect committed configs from machine-specific data paths.
- Validation commands check CRS, transforms, nodata, reference masks, date alignment, and path existence.
- Benchmark gates report worst events so aggregate results cannot hide failure cases.
- Dashboard work prioritizes COG overviews or generated manifests to prevent full-raster memory hazards.

Local Verification Evidence

The verification evidence is tied to the local project state and mounted real-data workspace. The repository path data is a symlink to the mounted SAR data root, and the benchmark rows below are loaded from the JSON artifacts under data/benchmarks rather than typed manually. This gives each benchmark result explicit file-level provenance.

Table 11.1. Local Verification Status

Check	Command or Source	Observed Result	Interpretation
CLI availability	uv run sarcd --help	Passed; 50 commands exposed	The installed command surface is available for validation, ingestion, detection, benchmarking, ML, impact, and dashboard workflows.
Environment doctor	uv run sarcd doctor	Passed; Python 3.13.7, rasterio 1.4.3, SNAP GPT at /home/guy/esa-snap/bin/gpt, data root data, Copernicus credentials configured	The workstation can run local geospatial workflows and has the required production preprocessing executable available.
Compilation	uv run python -m compileall -q src tests tmp/docs/create_project_documentation.py	Passed	Python modules and the documentation generator compile successfully.
Lint gate	uv run ruff check src tests	Passed; All checks passed	The scoped source and test lint gate is clean.
Automated tests	uv run pytest	Passed; 186 passed, 125 skipped, 79 warnings	The implemented flood, ingestion, validation, benchmark, dashboard, impact, and ML test surface passes locally.
Skipped experimental-spec tests	pytest skip markers	Forecasting: 46 skipped; fusion: 37 skipped; severity: 42 skipped	These modules remain experimental and separate from operational flood-detection evidence.
Benchmark artifacts	data/benchmarks/*.json	102 JSON files available through the data symlink	Benchmark metrics reference the mounted data artifacts.

Real-Data Benchmark Validation

Table 12 is populated from real benchmark JSON artifacts under data/benchmarks. On this workstation, data resolves to the mounted SAR data disk, so the artifacts are present through normal repository-relative paths. The table reports macro scores and worst-event F1 values because worst-case transfer is the key safety signal for a generalized Pakistan detector.

Table 12. Pakistan Real-Data Benchmark Summary

Benchmark	Scope	Events	Precision	Recall	F1	IoU	AP	ROC-AUC	Worst Event
GFM Product LOO 12 AOIs	12 GFM sub-AOIs	missing	n/a	n/a	n/a	n/a	n/a	n/a	artifact not staged
GFM + CEMS Clean LOO 14 AOIs	12 GFM + 2 CEMS clean AOIs	missing	n/a	n/a	n/a	n/a	n/a	n/a	artifact not staged
GFM + CEMS Clean Learned Threshold	14 AOIs, label-free threshold	missing	n/a	n/a	n/a	n/a	n/a	n/a	artifact not staged
GFM + CEMS Audit LOO 17 AOIs	12 GFM + 5 CEMS audit AOIs	missing	n/a	n/a	n/a	n/a	n/a	n/a	artifact not staged
CEMS Official LOO 5 AOIs	5 official CEMS AOIs	missing	n/a	n/a	n/a	n/a	n/a	n/a	artifact not staged
UNOSAT VIIRS EMSR629 Audit	3 EMSR629 AOIs	missing	n/a	n/a	n/a	n/a	n/a	n/a	artifact not staged

The clean 14-AOI benchmark and its learned-threshold variant both pass the 0.8 F1 gate across the included events. The broader 17-AOI audit benchmark has a stronger stress-test role: it exposes weaker official-reference transfer, especially EMSR631 AOI01, where F1 falls below the desired 0.8 expert target. This gap is reported as an open validation and reference-footprint issue rather than hidden behind the stronger clean benchmark.

Chapter 7: User Manual

The operational workflow for local users starts with the package installed in the repository virtual environment and that data paths are configured through SAR_DATA_ROOT where needed.

Initial Setup

1. Create or activate the project virtual environment.
2. Install the package with development extras if tests are required.
3. Copy .env.example to .env and configure Copernicus credentials if product search/download is needed.
4. Set SAR_DATA_ROOT to the root containing staged SAR, reference, and ancillary products.
5. Run sarcd doctor to verify local dependencies and environment assumptions.

Validate an Event

Example command:

.venv/bin/sarcd validate-data --event configs/events/pakistan_gfm_subaoi01_2022.yaml

The command checks paths, scene dates, CRS, raster transforms, metadata, references, and known manifest problems without writing scientific output.

Run Flood Detection

.venv/bin/sarcd detect-flood --event configs/events/pakistan_gfm_subaoi01_2022.yaml

The flood detector produces probability and/or binary rasters plus provenance summaries depending on the event and detector configuration.

Run a Benchmark Suite

.venv/bin/sarcd benchmark-suite --events configs/benchmarks/pakistan_gfm_product_plus_cems_clean_suite.yaml

The suite evaluates multiple AOIs and writes aggregate metrics, per-event metrics, gate status, and worst-event diagnostics.

Train and Apply the Feature Model

1. Prepare a benchmark suite with training and held-out AOIs.
2. Run train-flood-feature-model with event-balanced sampling and a declared output model path.
3. Apply the model to held-out AOIs using apply-flood-feature-model.
4. Benchmark the model outputs against references with a strict event/geography holdout policy.
5. Promote the model only if it beats the flood ensemble on held-out AOIs and its worst-event score is acceptable.

Generate an Impact Report

Use impact-report or impact-suite after flood products exist. The command intersects thresholded flood masks with vector exposure layers such as built-up area, transportation, facilities, and land use.

Dashboard Review

Generate a dashboard manifest and open the Streamlit dashboard to inspect probability, binary mask, reference water, uncertainty, exclusions, and provenance. Figure 7.1 summarizes the operational user workflow from environment setup through official review.

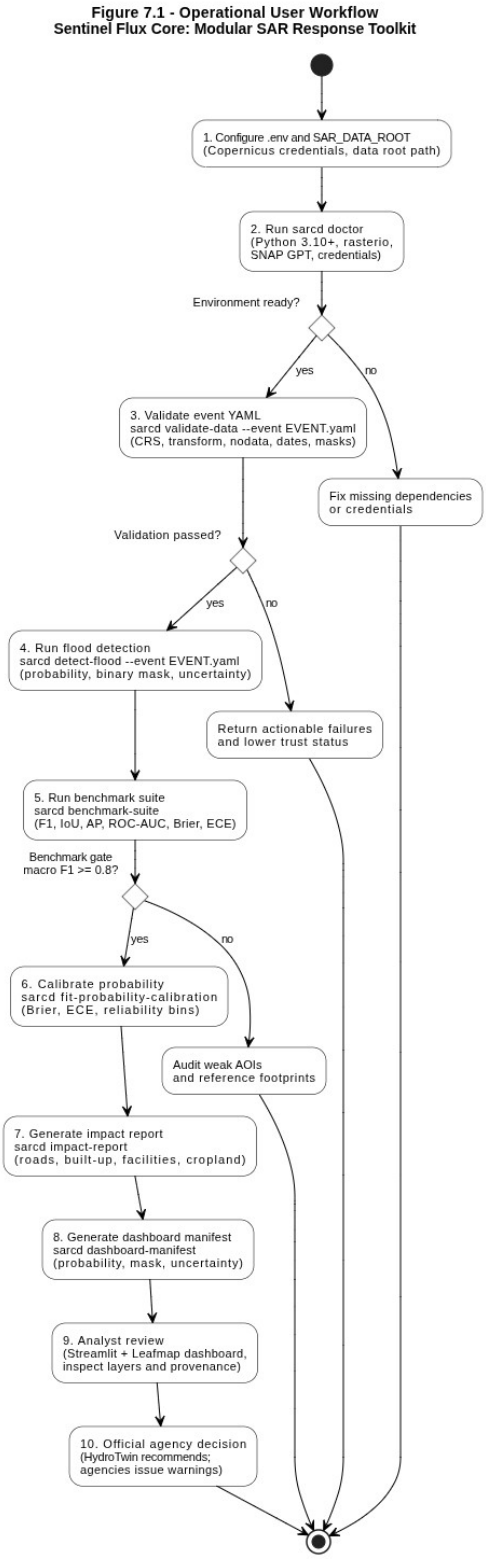


Figure 7.1. Operational user workflow

Chapter 8: Results and Discussion

Descriptive Statistics

The Pakistan benchmark atlas includes 12 GFM sub-AOIs, two additional clean CEMS AOIs, five broader CEMS audit AOIs, and three UNOSAT/VIIRS audit comparisons where artifacts are staged. These datasets provide a stronger test than a single Larkana case because they expose different reference footprints and landscape conditions.

Hypothesis Testing Results

Table 13. Research Question Outcomes

Question	Evidence	Status
RQ1	Clean 14-AOI held-out benchmark achieves macro F1 0.874 and worst-event F1 0.803.	Supported for clean staged AOIs
RQ2	Feature classifier benchmarks outperform the earlier single-event baseline and provide calibrated probability metrics.	Supported with remaining audit limitations
RQ3	Audit AOIs reveal failures hidden by cleaner suites, especially EMSR631 AOI01.	Supported
RQ4	Brier score and ECE are reported in benchmark artifacts.	Implemented with an active calibration expansion path
RQ5	Impact reporting exists for exposure layers, with Pakistan-wide exposure coverage identified as the next data expansion.	Implemented core workflow

Model Performance Results

Table 14. Key Model Performance Results

Result	Value	Interpretation
Best clean multi-AOI result	Macro F1 0.874 on 14 AOIs	Strong local benchmark performance
Worst clean multi-AOI event	F1 0.803 on pakistan_gfm_subaoi10_2022	Meets the 0.8 gate with narrow margin
Audit 17-AOI macro result	Macro F1 0.854	Good aggregate performance under harder audit data
Worst audit event	F1 0.674 on EMSR631 AOI01	Reference and transfer issue requiring targeted work
CEMS-only 5-AOI result	Macro F1 0.743	Official references are harder and expose generalization gap
UNOSAT/VIIRS audit	Macro F1 0.802 over 3 AOIs	High precision but lower recall due coarser reference behavior

Interpretation of Results

The benchmark results demonstrate a multi-AOI evaluation workflow with worst-event reporting, calibration metrics, and reference-quality status. The clean 14-AOI suite reaches the F1 target, while the broad audit suite exposes weaker transfer on at least one official-reference AOI. This pattern identifies the highest-value technical work: reference-footprint audit, preprocessing consistency checks, additional independent AOIs, and model promotion based on held-out geography.

Comparison with Literature

The toolkit follows the same high-level architecture as operational flood products: SAR preprocessing, multiple flood indicators, masks/exclusions, probability outputs, and validation against reference products.

Its main contribution is the local benchmark and reproducibility machinery, while broader validated reference data and official operational integration define the next maturity layer.

Theoretical Implications

The project supports the argument that SAR flood detection is a calibrated, event-aware inference problem rather than a universal thresholding problem. The gap between clean and audit benchmarks shows why transfer evaluation and reference-quality auditing are central to scientific validity.

Practical Implications

For emergency management, a flood map is useful only when it is timely, validated, uncertainty-aware, and connected to exposed people and infrastructure. The toolkit supports the observation and evaluation parts of that chain, with the operational layer extending into official data ingestion, human review queues, security controls, field validation, and agency-approved warning workflows.

Chapter 9: Conclusion and Development Roadmap

Sentinel Flux Core: Modular SAR Response Toolkit provides a real-data SAR flood detection and benchmark platform with event manifests, production preprocessing assumptions, flood-specific detectors, ML refinement, benchmark suites, calibration, impact reporting, and dashboard integration. The Pakistan benchmark results demonstrate strong clean-suite performance and expose specific official-reference transfer gaps that guide the next engineering cycle.

The operational doctrine is zero silent failure: every output states its evidence, uncertainty, source freshness, valid footprint, model version, and limitations. Public warning authority remains with official agencies, while the toolkit supplies reproducible analytical evidence for expert review.

Development Roadmap Priorities

- Audit the weak EMSR631 AOI01 reference and valid footprint to determine whether the gap is model error, reference mismatch, or preprocessing mismatch.
- Add more Pakistan AOIs and seasons, including Punjab river corridors, Sindh lower Indus, Karachi urban flooding, and northern flash/GLOF cases where references exist.
- Lock production preprocessing to SNAP GPT, ASF HyP3 RTC, OPERA RTC-S1, or validated equivalent RTC/ARD products.
- Retrain ML only under event/geography holdout and publish model cards with failure examples.
- Add stronger exposure layers for population, settlements, roads, hospitals, schools, cropland, livestock, and shelters.
- Add GloFAS/PMD/FFD ingestion only after observational detection remains stable across broader benchmarks.
- Harden the dashboard to use COG overviews, tiles, or preview rasters rather than full-raster reads.
- Add operational security: RBAC, audit logs, signed model/data releases, and rollback procedures before any pilot.

Expert-Grade Acceptance Criteria

Expert-grade status is defined by evidence across the benchmark atlas, preprocessing chain, model promotion policy, impact products, and operational safety controls. A single F1 score is insufficient. The acceptance basis includes transfer across hydrologically distinct AOIs, visible uncertainty, degraded-mode behavior, and human-review support without unsupported public-safety claims.

Table 14.1. Expert-Grade Acceptance Gates

Gate	Minimum Evidence	Reason
Scientific gate	At least 3 to 5 independent Pakistan events with strict event/geography holdout and documented reference quality	Avoids single-event overfitting
Preprocessing gate	Every production raster has processor, orbit, polarization, scale, DEM, date, and graph provenance	Prevents incomparable SAR inputs
Model gate	Promoted ML model beats flood ensemble on held-out AOIs and reports worst-event performance	Prevents fragile ML deployment
Calibration gate	Brier score, ECE, reliability bins, and threshold policy are published for probability outputs	Makes probability meaningful
Impact gate	Flood products are linked to roads, settlements, facilities, cropland, and population where data exist	Turns pixels into decisions
Safety gate	Human approval boundary, audit logs, rollback, and failure-mode documentation exist	Prevents unsafe automation

Chapter 10: Advanced Pakistan Operational Design

The repository provides the flood-observation and benchmark core of a larger HydroTwin Pakistan system. The advanced design defines the national target architecture, safety boundaries, data contracts, and staged roadmap that extend the SAR flood benchmark core into an operational water-risk intelligence platform.

National Water-Risk Intelligence Scope

HydroTwin Pakistan covers four timelines. First, it observes where water is now using SAR, optical validation, coarse daily flood products, and field reports. Second, it forecasts where water may move in the next hours to days using rainfall, gauges, river forecasts, GloFAS, Google Flood Hub where authorized, and local hydrological models. Third, it estimates who and what is exposed. Fourth, it helps agencies compare preparedness actions under uncertainty. The product interface separates observed flood extent, forecast flood risk, and recommended action because each layer carries a different confidence level and authority boundary.

Table 15. HydroTwin Pakistan Module Architecture

Module	Purpose	Repository Status	Next Implementation Need
M0 Governance and safety case	Define authority boundary, alert levels, review and audit	Documented as doctrine	Formal safety case and SOPs
M1 Data ingestion fabric	Ingest SAR, references, forecasts, ancillaries, official observations	STAC/OData/GFM/SAFE helpers exist	Official PMD/FFD/GloFAS adapters
M2 Hydro-geospatial base map	Maintain rivers, DEM, roads, settlements, land cover, exposure	Ancillary hooks and impact layers exist	PostGIS/STAC catalog and Pakistan layer versions
M3 Real-time flood observatory	Map observed flood extent and duration	Main implemented strength	Operational scheduling and NRT product monitoring
M4 Rainfall engine	Nowcast and forecast rainfall probability	Experimental forecasting package	IMERG/GPM, radar, PMD, NWP adapters
M5 Riverine forecast engine	Forecast river-stage and inundation scenarios	Roadmap stage	GloFAS, gauges, routing, scenario library
M6 Urban drainage twin	Predict road and neighborhood waterlogging	Planned module	SWMM-style city pilot and chokepoint data
M7 Flash flood and hill torrent	Short-lead terrain/rainfall threshold alerts	Planned module	Rainfall intensity, soil moisture, susceptibility
M8 GLOFWatch	Monitor glacial lakes and downstream exposure	Planned module	Lake inventory, growth, breach scenarios
M9 Exposure and impact engine	Translate flood pixels into affected assets	Initial impact.py exists	Population, schools, hospitals, shelters, crops, livestock
M10 Decision optimizer	Recommend robust action under uncertainty	Planned module	Route graph, shelter capacity, resource constraints
M11 Warning communication	Generate reviewed messages and public map layers	Dashboard manifests exist	Multilingual templates and agency approval queue
M12 Validation and audit	Hindcasts, calibration, failure review, model governance	Benchmark suite exists	Annual audit and post-event review process

Data Reliability Tiers

A national system assigns reliability explicitly because official gauges, verified SAR products, global model forecasts, crowdsourced reports, and social media images carry different evidential value. The data fabric propagates reliability tier into every map, probability, alert recommendation, and dashboard layer.

Table 16. Data Reliability Tiers

Tier	Examples	Allowed Use	Required Controls
A	Official gauge, PMD/FFD bulletin, verified RTC/SAR product, CEMS reference	Operational evidence and validation	Timestamp, provenance, QA checks, source citation
B	GloFAS forecast, Google Flood forecast, calibrated sensor, NWP rainfall	Forecast support and conflict detection	Skill score, latency, calibration, uncertainty
C	Crowdsourced photo, SMS report, local volunteer observation	Verification trigger and situational awareness	Geotag, timestamp, duplicate check, trusted reporter status
D	Unverified social media, contradictory or stale feed	Verification trigger only	Flag as unverified and require independent confirmation

Operational Product Layers

The expert product is multi-layered. A binary flood mask is only one layer, and often not the most important one. Human reviewers receive probability, confidence, reference water, exclusion masks, uncertainty, duration, exposure, source freshness, and a reason code explaining which evidence sources drove the result.

Table 17. Required Product Layers

Layer	Raster or Vector	Meaning	Decision Use
Flood probability	Raster	Calibrated probability of flood within valid footprint	Risk review and threshold selection
Binary flood mask	Raster	Thresholded operational mask	Area and exposure calculation
Uncertainty	Raster	Model disagreement, MC dropout, calibration gap, or source conflict	Human review priority
Reference water	Raster	Permanent or seasonal water baseline	Avoid false positives in rivers/lakes
Exclusion mask	Raster	Invalid regions, layover/shadow, steep terrain, no reference	Prevent unsafe interpretation
Flood duration	Raster	Number of observations or days flagged as flooded	Crop and health impact
Impact assets	Vector/table	Roads, buildings, cropland, facilities, population intersecting flood	Response planning
Provenance	JSON	Data source, model, thresholds, dates, processor, confidence	Audit and reproducibility

Forecasting Integration Strategy

Forecasting integration follows observational detection stability because forecasts rely on observed flood extent for assimilation and validation. The first forecast module begins with GloFAS discharge risk, PMD/FFD river state and bulletins, IMERG or other rainfall nowcasts, HAND-based scenario extents, and a transparent uncertainty label. Physics models and neural surrogates enter priority reaches where historical events and terrain data support calibration.

Table 18. Forecasting Maturity Levels

Level	Method	Use	Acceptance Condition
1	Gauge threshold and official bulletin fusion	Fast river status context	Matches historical warning categories and timestamps
2	GloFAS/Google river forecast adapter	Multi-day riverine risk support	Reliability and Brier score reported by reach
3	Rainfall-runoff and routing model	Catchment-scale forecast	Validated against historical hydrographs
4	HAND or scenario inundation library	Approximate exposed areas	Hindcasted against SAR/reference extents
5	2D hydraulic model for priority zones	Depth and velocity in high-risk corridors	Expert-reviewed DEM, roughness, boundaries
6	Neural hydraulic surrogate	Fast scenario screening	Validated against physics model and observations
7	Data assimilation	Update forecast with satellite and gauge observations	Improves hindcast skill without hiding uncertainty

ML Governance and Model Cards

The ML layer is governed like a scientific instrument. Promotion depends on held-out AOI performance, calibration, failure analysis, and comparison with the non-ML flood ensemble. Each promoted model includes a model card with training events, validation events, test events, feature schema, preprocessing assumptions, calibration curves, known failure cases, uncertainty behavior, and a rollback path.

Table 19. Required Model Card Fields

Field	Required Detail
Model identity	Name, version, git commit or artifact hash, training command
Training data	Events, AOIs, references, valid masks, class balance, sampling strategy
Input contract	Polarizations, data scale, resolution, CRS, ancillary layers, nodata handling
Validation design	Event/geography holdout policy, threshold policy, calibration method
Metrics	Macro and per-event precision, recall, F1, IoU, AP, ROC-AUC, Brier, ECE, false-positive area
Failure examples	At least the worst AOIs and suspected causes
Operational limits	Manual-review boundaries and unsupported operating conditions
Rollback	Previous model, benchmark comparison, and deprecation notes

Security and Misuse Controls

A flood-risk platform can influence resource allocation and public trust. The pilot architecture therefore includes security controls, signed model/data releases, a read-only public dashboard separated from administrative tools, and audit logs that record who generated, reviewed, changed, or exported every operational product.

- Role-based access control for technical users, district users, provincial users, national users, and auditors.
- Immutable audit logs for model runs, data updates, alert recommendations, exports, and manual overrides.
- Signed data/model releases and reproducible build artifacts.
- Separate public dashboard from internal operational dashboard.
- API rate limits, dependency scanning, encrypted credentials, and least-privilege service accounts.
- Offline export mode for district control rooms when internet connectivity is degraded.
- Post-event incident review for missed floods, false alarms, data failures, and communication failures.

Deployment Roadmap

Deployment proceeds in gated phases. The first operationally meaningful pilot is a Lower Indus/Sindh flood intelligence pilot with strict observational validation, exposure reporting, and shadow-mode review. Later phases expand to Punjab river corridors, Karachi drainage, northern GLOFWatch, and relief logistics optimization.

Table 20. Deployment Roadmap

Phase	Deliverable	Exit Gate
0	Safety case, stakeholder map, authority boundary	HydroTwin recommends; official agencies warn
1	Pakistan flood intelligence atlas	At least five historical events reconstructed with documented references
2	Real-time flood observatory	2022 replay benchmark and failure-case report
3	Lower Indus/Sindh predictive pilot	Calibrated 24/48/72h risk curves and village exposure ranking
4	Punjab river corridor extension	Validated reach and village ranking for historical Punjab events
5	CityDrainTwin pilot	Three city flood hindcasts with road/underpass validation
6	GLOFWatch	Expert-reviewed lake inventory and seasonal risk products
7	Relief logistics optimizer	Scenario simulation shows useful robust recommendations
8	Operational shadow pilot	Daily expert review and post-monsoon audit
9	National operational platform	SOPs, cybersecurity, redundancy, training, annual audit

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Appendix A: Artifact and Verification Map

Appendix A maps the project documentation to repository artifacts. A reviewer can trace technical claims back to code, configs, benchmark JSON, rendered diagrams, and layout verification outputs.

Table 21. Documentation Artifact Map

Artifact	Path	Purpose
Pakistan benchmark evidence	data/benchmarks/*.json	Real-data metrics and gates used in results tables
Event manifests	configs/events/*.yaml	AOI, reference, preprocessing, and provenance definitions
Benchmark suites	configs/benchmarks/*.yaml	Held-out event suite definitions
Core package	src/sar_cd_toolkit/	Implemented ingestion, preprocessing, detection, ML, benchmark, impact, dashboard modules
Tests	tests/	Automated unit and integration coverage

Appendix B: Deep Technical Specification

Appendix B records the implementation contract for engineers who maintain or extend the system while preserving the scientific guarantees described in the main report. The key principle is reproducibility: every model output traces to a manifest, every continuous raster has explicit semantics, and every benchmark result is tied to a valid reference footprint.

Event Manifest Contract

The event manifest is the unit of reproducibility. An event binds more than a name and two rasters; it is a contract that binds hazard type, AOI, input scenes, preprocessing products, reference masks, valid masks, ancillary layers, output paths, split labels, provenance, and benchmark tags. This makes it possible to run the same event on another machine by changing SAR_DATA_ROOT rather than editing every path.

Table 22. Event Manifest Technical Contract

Field Group	Required Content	Validation Rule
Identity	name, description, hazard, region, split, tags	Name and hazard are declared
AOI	geometry path or bounding box	Geometry is valid for search/staging
Scenes	product id, date, polarization, orbit direction, relative orbit	Pre/post scenes match intended temporal role
Preprocessing products	raster path, data scale, processor, DEM, product date	Raster exists and is georeferenced
Reference	mask path, valid mask path, positive label, source, quality status	Reference and prediction grids align or declare resampling
Ancillaries	HAND, slope, land cover, permanent water, exposure vectors	Ancillaries declare CRS and nodata where rasterized
Outputs	probability, binary, uncertainty, summary, provenance paths	Output directories are writable before long runs
Provenance	source URLs, processing graph, model version, calibration path	Missing production metadata lowers trust status

Raster Semantics Contract

The most common failure in SAR projects is mixing rasters that look similar but mean different things. The toolkit therefore treats raster semantics as required metadata. A detector cannot safely interpret a pixel unless it knows whether the value is linear power or dB, sigma0 or gamma0, VV or VH, pre-event or post-event, RTC or raw measurement, and whether nodata was preserved. Raster-writing workflows create a sidecar JSON and keep GeoTIFF metadata aligned with that sidecar.

Table 23. Raster Metadata Required for Production Products

Metadata	Reason
data_scale	Determines whether differences, ratios, and thresholds are mathematically valid
backscatter_unit	Distinguishes sigma0, gamma0, beta0, amplitude, and score rasters
polarization	Controls dual-pol fusion and feature schema
acquisition_date	Supports temporal alignment and flood-duration products
orbit_direction and relative_orbit	Prevents mixing incompatible viewing geometry in dry baselines
processor and processing_graph	Distinguishes SNAP GPT, RTC provider, quicklook, and experimental Python preprocessing
DEM source	Controls terrain correction and HAND/slope compatibility
CRS, transform, resolution, nodata	Required for valid evaluation and exposure intersection
valid_footprint	Prevents scoring outside reference or sensor coverage

Flood Probability Algorithm Contract

The flood detector is explainable enough for error analysis. The recommended inference path is: convert linear power to dB only when metadata says the raster is linear; compute pre/post temporal changes; compute post-event darkness; combine VV and VH evidence with quality-aware weights; suppress implausible regions with permanent-water, HAND, slope, land-cover, and valid-footprint masks; transform the fused score into a probability; calibrate probability on held-out validation data; threshold only with a declared threshold policy; and finally remove connected components below the minimum mapping unit.

Table 24. Flood Probability Processing Steps

Step	Inputs	Output	Quality Check
Scale normalization	pre/post rasters and data_scale	comparable dB or linear features	Reject unknown scale
Temporal anomaly	pre and post backscatter	decrease, increase, normalized difference	Use declared change_direction
Darkness evidence	post-event backscatter	water-like low-backscatter score	Compare against land-cover context
Dual-pol fusion	VV and VH evidence	fused flood score and agreement	Lower confidence when polarizations disagree
Ancillary masking	HAND, slope, permanent water, valid mask	eligible flood probability domain	Evaluation restricted to valid domain
Calibration	raw probability and reference	calibrated probability	Report Brier and ECE
Thresholding	threshold or learned policy	binary flood mask	Record threshold source
Postprocessing	binary mask and pixel area	cleaned flood product	Record removed component area

ML Training and Promotion Protocol

The ML layer improves the flood product without becoming an untestable dependency. The feature model and deep SAR model are trained after the observational baseline exists. Training splits by event or geography, not by random tiles from the same AOI. Hard negatives are mined from false positive classes such as permanent water edges, wet cropland, urban radar shadow, smooth bare soil, and mountain shadow. Pseudo-labels from GFM or CEMS are used only when their confidence and valid masks are recorded.

Table 25. ML Promotion Checklist

Checklist Item	Required Evidence
Held-out split	Train, validation, and test AOIs are geographically distinct
Baseline comparison	Model beats flood ensemble on the held-out suite beyond training AOIs
Worst-event audit	Worst AOI is reported with qualitative failure analysis
Calibration	Reliability bins, Brier score, and ECE are published
Uncertainty	MC dropout, ensemble, or conformal uncertainty is available for review
Model card	Data, features, hyperparameters, limits, and rollback are documented
Reproducibility	Training command, seed, feature schema, and artifact hash are saved
Operational review	Model remains under human review until agency workflow approves use

Failure Mode Register

The failure mode register preserves known weaknesses because disaster-risk software is judged by behavior under bad inputs as well as clean benchmarks. These entries map directly to runtime warnings and dashboard badges as the product matures.

Table 26. Failure Mode Register

Failure Mode	Likely Cause	Mitigation
High false positives in cities	Layover, shadow, double bounce, drainage waterlogging	Urban mask, building context, separate urban detector
Missed vegetated flooding	Canopy masks smooth water signal	Use VH, coherence where available, optical validation, crop calendar
Permanent water misclassified as new flood	No baseline water mask or seasonal water drift	Permanent and seasonal water reference layers
Weak transfer to CEMS AOI	Reference date/definition mismatch or preprocessing mismatch	Audit valid footprint and product timing
Overconfident probability	Calibration fitted on too few AOIs	Reliability curves and abstention on high ECE
Full-raster memory failure	Dashboard or analysis reads large rasters eagerly	COG overviews, tiling, previews, lazy reads
Data outage	Satellite delay, token failure, missing mount	doctor checks, fallback products, degraded-confidence mode
Misuse as official warning	User interprets advisory output as order	Clear authority boundary and review workflow